

PHY 121
ELECTRICITY, MAGNETISM and MODERN PHYSICS

- RESISTANCE IN SERIES AND PARALLEL
- E.M.F.
- KIRCHOFF'S LAWS
- POTENTIOMETER
- WHEATSTONE BRIDGES

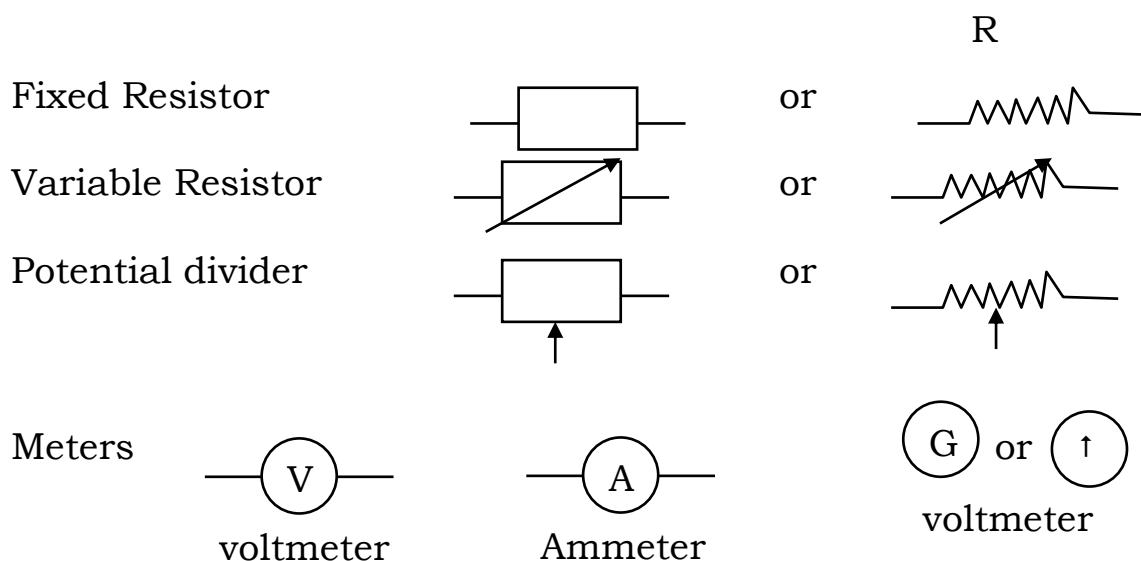
Prepared By
OGBAN PETER U.

RESISTANCE

The resistance R of a conductor is defined as the ratio V/I , where V is the potential difference V is applied to two conductors A and B , and a smaller current I flows in A , then the resistance of A is greater than that of B . we then write:

$$\frac{V}{I} = R$$

Symbols used to Represent Different Type of Resistors and Meters



Conductance (a): is defined as the ratio I/V and is therefore, the inverse of resistance (i.e. I/R in numerical value). The unit of conductance is Siemens (s).

Example:

- (1) A current of 2 mA flows in a ratio resistance R when a potential difference of 4v is connected. What are the values of resistance and conductance?

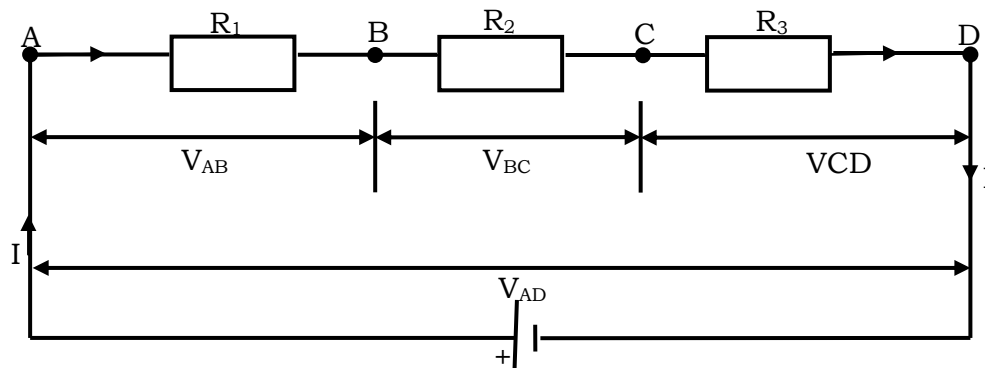
Solution:

$$\text{Recall, } R = \frac{V}{I} = \frac{4}{(2 \times 10^{-3})} = 2 \times 10^3 = 2000 \Omega$$

Resistance in Series

The resistors of an electric circuit may be arranged in series, so that the charges carrying the current flow through each in turn, or they may be arranged in parallel, so that the flow of charge divides between them.

Consider three passive resistors in series, carrying a current I , as shown in the diagram below:

***Resistance in Series***

If V_{AD} is the potential difference across the whole system, the electrical energy supplied to the system per second is IV_{AD} electrical energy per second in all the resistors.

Therefore, $IV_{AD} = IV_{AB} + IV_{BC} + IV_{CD}$

$$\Rightarrow V_{AD} = V_{AB} + V_{BC} + V_{CD} \quad \dots\dots\dots (1)$$

the individual potential difference are given as:

$$V_{AB} = IR_1, V_{BC} = IR_2, V_{CD} = IR_3 \quad \dots\dots\dots (2)$$

From equ. 1,

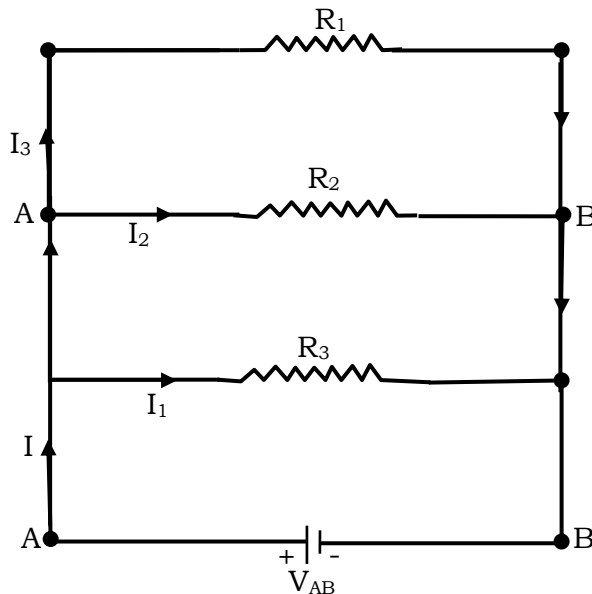
$$\begin{aligned} V_{AD} &= IR_1 + IR_2 + IR_3 \\ &= I(R_1 + R_2 + R_3) \quad \dots\dots\dots (3) \end{aligned}$$

Also, the effective resistance of the system,

$$R = \frac{V_{AD}}{I} = R_1 + R_2 + R_3 \quad \dots\dots\dots (4)$$

Resistance in Parallel

Consider there passive resistors connected in parallel between the points A, B as shown in the figure below,



The potential difference between A and B is the same for each resistor. A current I enters the system at A and leaves at B setting up a potential difference V_{AB} between those points. The current branches into I_1, I_2, I_3 through the three elements therefore, $I = I_1 + I_2 + I_3$.

$$\Rightarrow I_1 = \frac{V_{AB}}{R_1}, I_2 = \frac{V_{AB}}{R_2}, I_3 = \frac{V_{AB}}{R_3}, \dots\dots\dots (5)$$

$$\Rightarrow I = V_{AB} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

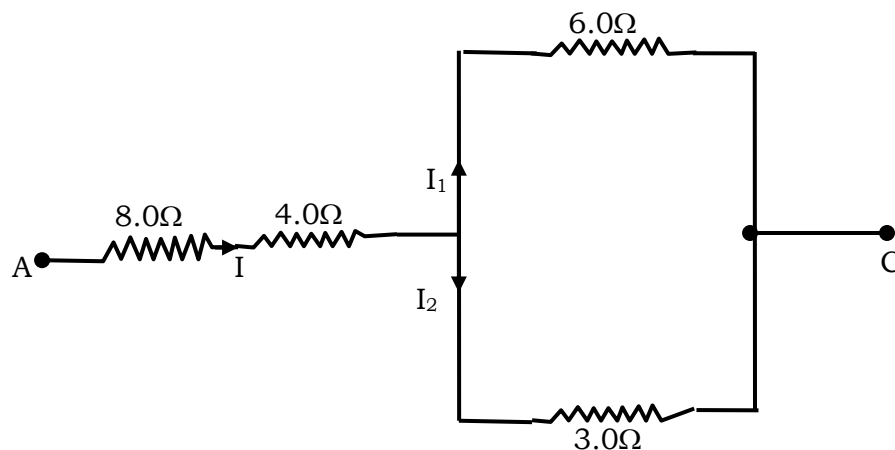
$$\text{Also, } \frac{I}{V_{AB}} = \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots\dots\dots (6)$$

Where, R is the effective resistance (V_{AB}/I) of the system.

Examples

Four resistors are connected as shown below

- Find the equivalent resistance between points A and C
- What is the current in each resistor if a potential difference of 42v is maintained between A and C.



Solution

The Equivalent resistance in series

$$R_{eq} = R_1 + R_2 = 8.0 + 4.0 = 12\Omega$$

The equivalent resistance in parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\Rightarrow R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{1}{\frac{1}{6} + \frac{1}{3}} = \frac{1}{\frac{1}{2}}$$

$$\therefore R_{eq} = 2.0\Omega$$

The equivalent resistance from A to C is

$$12 + 2 = 14\Omega$$

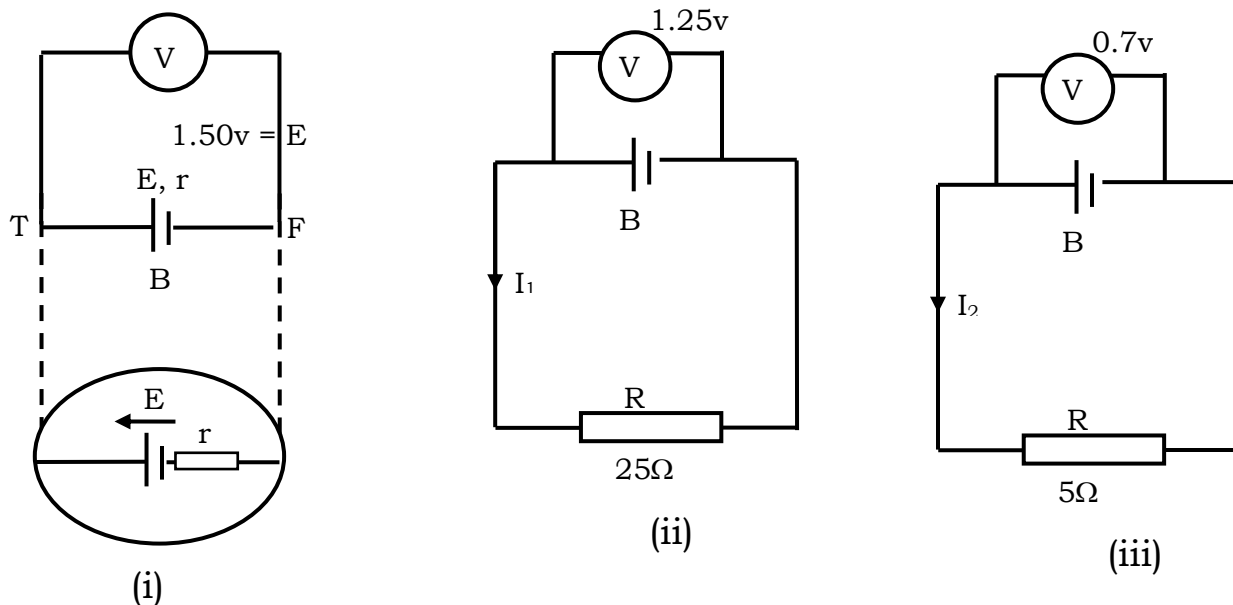
(b) Using $R = \frac{V_{AB}}{I}$

$$\therefore I = \frac{V_{AC}}{R_{eq}} = \frac{42V}{14\Omega} = 3.0A$$

Electromotive Force (E.M.F)

Consider the current and potential difference P.d., in circuits connected to a generator such as a battery. If a high resistance voltmeter is connected across the terminals of a dry battery B, the meter may read about 1.5v, since practically no current flows from the battery in this case we say it is an 'open circuit' the p.d. across the terminals of a battery (or any other generator) on open circuit is called ELECTROMOTIVE FORCE or EMF (E).

When a resistor is connected to the battery, the current flows through the internal resistance, r , of the battery to complete the circuit flow. The e.m.f. of a battery depends on the nature of the chemicals used and not on its size. A tiny battery has the same e.m.f. as a large battery made of the same chemicals. The internal resistance of the tiny battery, however, is much less than that of the large battery. Provided only a small current is taken from the battery, its e.m.f. and resistance are fairly constant.



E.m.f. and internal resistance

Any electrical generator has two important properties, an e.m.f; E and an internal resistance r . As shown in figure (i) above E , and r may be represented separately in a diagram, though in practice there are together between the terminals T.T. so we can think of the battery as an electric pump with its e.m.f. E pushing the current

round the circuit through both the external (outside) resistor R and the internal resistor r . It implies that the e.m.f. E is the p.d. across the external resistor R plus the p.d. across the internal resistance r .

Terminal P.D

In figure (ii) above, a resistor of 25Ω is connected to the battery B so that a current I , flows in the circuit. The voltmeter reading across the battery terminals, or terminal p.d., may then be 1.25v , although the e.m.f. is 1.5v . When the resistor is replaced by one of 5Ω as in figure (iii), a larger current I_2 flows and the voltmeter reading or terminal p.d. is now 0.7v . The terminal p.d. varies when a current flows from a battery, because the voltmeter is connected across the external or outside resistance in figure (ii). So 1.25v is the p.d. across the 25Ω resistor. Now the e.m.f., 1.5v maintains the current in the whole circuit, that is through the external and internal r . therefore deduce that the p.d. across the internal resistance $r = 1.5 - 1.25 = 0.25\text{v}$.

Similarly, in figure (iii), 0.75v is the p.d. across the external resistance 5Ω . So in this case the P.d. across the internal resistance $r = 1.5 - 0.75 = 0.75\text{v}$. A common error is to think that the voltmeter across the terminals reads the e.m.f. This is not the case here as there is a P.d. across the internal resistance when a current flows, and the voltmeter can only read the p.d. across the external resistance R , which is the terminal p.d.

In figure (ii), the p.d. across the 25Ω external resistor R is 1.25v and the p.d. across the internal resistance $r = 0.25\text{v}$. Since the same current flows in R and r it flows that $R = 5r$ or $r = R/5 = 5\Omega$.

Similarly, the p.d. across the external resistor R of 5Ω in figure (iii) is 0.75v and that across the internal resistance r is 0.75v . so $r = R = 5\Omega$, as previously calculated.

Note that as the external resistance R increases, the terminal p.d increases. When R is an infinitely high value, so that $I = 0$, the terminal p.d. is equal to the e.m.f. It therefore implies that e.m.f., $E = \text{p.d across the whole circuit, } R \text{ plus } r$.

From figure (ii), the p.d. across R is IR and the p.d across r is Ir . So,

$$E = IR + Ir = I (R + r) \quad \text{.....} \quad (1)$$

$$\text{Or } I = \frac{E}{R+r} \quad \text{.....} \quad (2)$$

Note that when the e.m.f., E is used to find the current I , the resistance $(R + r)$ of the whole circuit is required. On the other hand, the terminal p.d., $v = \text{p.d. across external resistor } R$

$$\Rightarrow \text{terminal p.d. } v = IR = \frac{ER}{R+r}$$

$$\text{Further, from (1) } V = E - Ir \quad \text{.....} \quad (3)$$

And the current from (1). $Ir = E - IR = E - V$

$$\therefore r = \frac{E-v}{I} \quad \text{.....} \quad (4)$$

Example

Suppose a current of 0.5A flows from a battery of an e.m.f., unknown and internal resistance 4Ω having a terminal p.d. of 1V. Calculate the e.m.f.

Using $V = E - Ir$

$$\Rightarrow E = V + Ir = 1 + (0.5 \times 4)$$

$$E = 3V$$

KIRCHOFF'S LAWS

Kirchoff extended Ohm's law to the network of electrical conductors and gave two laws, which together enabled the current in any part of the network to be calculated.

Kirchoff's first law states that the total current flowing into the junction in a network is equal to the total current flowing out of it.

Consider the figure below:

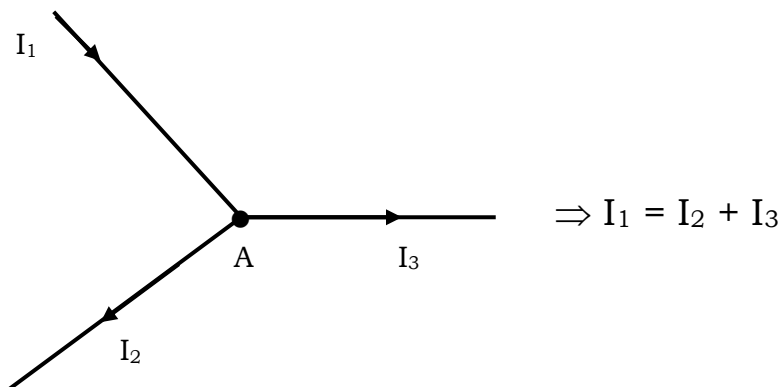


Fig. (i)

The algebraic sum of the current at a junction of a circuit is zero. Where a current is assigned positive if it flows towards the point, and negative if it flows away from it, it implies that at point A in figure (i) $I_1 - I_2 - I_3 = 0$

Consider the figure below:

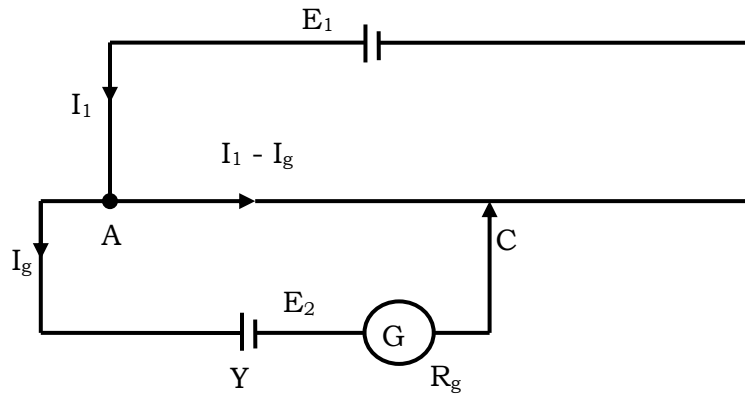


Fig. (ii)

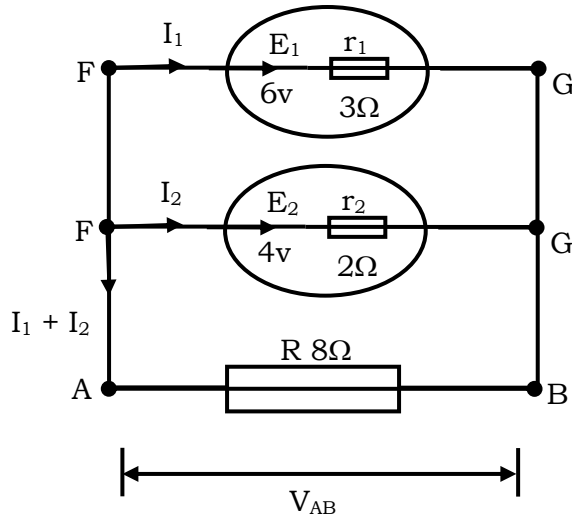
Kirchoff's second law states that; in a closed loop, the algebraic sum of the e.m.f.s is equal to the algebraic sum of all the p.d.s in that circuit.

So, going clockwise round the loop AYCA in fig. (ii) above,

$$E_2 = R_{AC} (I_1 - I_g) - R_g I_g$$

Example

Using Kirchoff's laws find the currents I_1 & I_2 from the network below, having tow batteries one of the e.m.f. 6v and internal resistance 3Ω and the other of e.m.f. 4v and internal resistance 2Ω are in parallel across an external resistance R of 8Ω



From Kirchoff's first law, the current passing through the 8Ω wire is $(I_1 + I_2)$, assuming I_1 and I_2 are the currents through the cells.

Taking closed circuits formed by each cell with the 8Ω wire, from the second law,

$$E_1 = 6 = 3I_1 + 8(I_1 + I_2) = 11I_1 + 8I_2$$

$$E_2 = 4 = 2I_2 + 8(I_1 + I_2) = 8I_1 + 10I_2$$

Solving the two equations

$$I_1 = 0.61\text{A}, I_2 = -0.09\text{A}$$

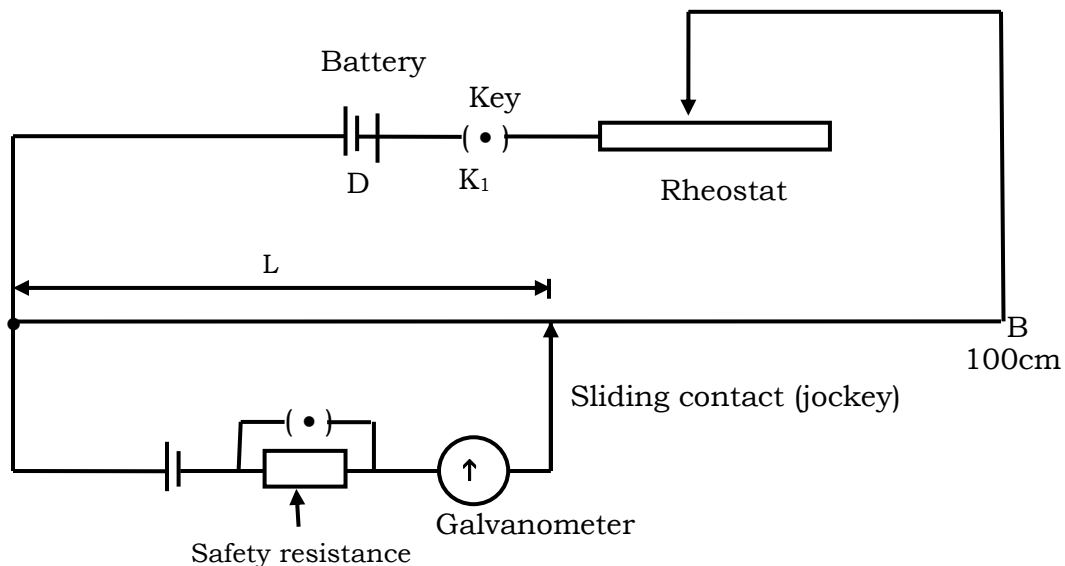
POTENTIOMETER

A potentiometer is a three-terminal resistor component with a sliding contact that forms an adjustable voltage divider. When two terminals are used, it acts as a variable resistor or rheostat. Potentiometer measuring instrument is essentially a voltage divider used for measuring electric potential (voltage); the component is an implementation of the same principle.

USES OF POTENTIOMETER

Potentiometers are commonly used to control electrical devices such as volume controls on audio equipment. Potentiometers operated by a mechanism can be used as position transducers, for example, in a joystick. Potentiometer would be comparable to the power in the controlled load.

When a steady current passes through a uniform resistance wire the potential gradient along it is also uniform. For example, if an electric current of 0.5A passes through a uniform wire of resistance 5 Ohms and one metre long, the resistance of each centimeter is 0.05 Ohms, the potential difference across each centimeter is 0.025 volts (i.e. $0.5\text{A} \times 0.05\text{ Ohms}$) using Ohm's Law.



As the wire is uniform, the uniform potential gradients along the wire are the potential difference across the whole length of the wire

divided by the length of the wire. The drops V_1 , V_2 across two different lengths, L_1 , L_2 of the same wire are respectively given by:

$$V_1 = 0.025L_1 \text{ and}$$

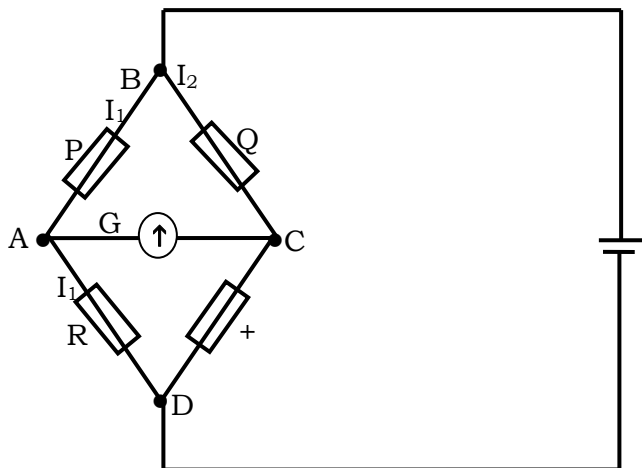
$$V_2 = 0.025L_2$$

So that, $\frac{V_1}{V_2} = \frac{L_1}{L_2}$

Thus, the potential drops down two different lengths'. This illustrates the method of obtaining potential differences of known values and measuring the ratio between them. The potentiometer operates on this principle.

WHEATSTONE BRIDGE CIRCUIT

Wheatstone designed a circuit called a 'bridge circuit' which gave an accurate method for measuring resistance.



In figure (i) above, x is the unknown resistance, and P, Q, R are resistance boxes. One of these usually R is adjusted until the galvanometer G between A and C shows no deflection, a so-called 'balance' condition. In this case the current I_g in

$$\frac{P}{Q} = \frac{R}{X}$$

$$\therefore X = \frac{QR}{P}$$

Wheatstone Bridge Proof

At balance, since no current flows through the galvanometer, the points A and C must be at the same potential.

Therefore, $V_{AB} = V_{CB}$ and $V_{AD} = V_{CD}$

$$\text{Using ratios, } \frac{V_{AB}}{V_{AD}} = \frac{V_{CB}}{V_{CD}} \quad \text{.....} \quad (i)$$

Since $I_g = 0$, P and R carry the same current, I_1 , and X and Q carry the same current, I_2 , so from

$$V = IR$$

$$\therefore \frac{V_{AB}}{V_{AD}} = \frac{I_1 P}{I_1 R} = \frac{P}{R}$$

$$\text{and } \frac{V_{CB}}{V_{CD}} = \frac{I_2 Q}{I_2 X} = \frac{Q}{X} \quad \text{.....} \quad (ii)$$

from eqn. (i) and (ii)

$$\frac{P}{R} = \frac{Q}{X}$$

$$X = \frac{QR}{P}$$

Exactly the same relationship between the four resistance is obtained if the galvanometer and cell positions are interchanged. Further analysis of the circuit shows that the bridge is most sensitive when the galvanometer is connected between the junction of the highest resistances and the junction of the lowest resistances.

COURSE:

ELECTRICITY AND MAGNETISM

- ❖ **ELECTROLYSIS**
- ❖ **FARADAY'S LAWS OF ELECTROLYSIS**
- ❖ **MAGNETIC FIELDS**
- ❖ **MAGNETIC FORCES ON CONDUCTORS**
- ❖ **HALL'S EFFECT**

By

ACHIBIRI SAMUEL UZOMA SMITH

ELECTROLYSIS

Electrolysis is the process of using direct current (DC) to initiate a chemical reaction. Electrolysis is derived from two Greek words – “electron” and “lysis” – meaning amber or change and dissolution. Knowledge and use of electrolysis has been known since 1785 when Martins Van Marum used electrolysis to reduce tin, zinc and antimony from their salts.

Key Words and Phrases

- ❖ **Anode:** This is the positive electrode in a cell.
- ❖ **Cathode:** This is the negative electrode in a cell.
- ❖ **Electrolysis:** The production of a chemical reaction by the flow of electric current through a solution.
- ❖ **Valence:** Represents the net change in electronic units carried by an ion.
- ❖ **Ion:** An ion may be either a charged atom or a charged group of atoms.
- ❖ **Electrolyte:** This is a substance containing free ions which are the carriers of electric current in electrolysis. If the ions are not mobile as in a solid salt, electrolysis cannot take place.
- ❖ **Chemical Equivalent Weight:** The atomic weight of an element divided by its valence is known as the chemical equivalent weight of the element.

- ❖ **Electrodes:** These are conductors in the form of wires, rods or plates through which an electric current enters or leaves the electrolyte.
- ❖ **Cations:** These are the positively charged ions in an electrolyte.
- ❖ **Anions:** These are the negatively charged ions in an electrolyte.

When an electrolyte is dissolved in water, it breaks up into positively and negatively charged ions, making the solution an electric conductor. When a potential difference is applied to the two electrodes, the battery or generator of electric current pumps electrons from its anode to the cathode. The negatively charged cathode now attracts cations in the electrolyte to itself. The cations accept electrons to become electrically neutral and are eventually discharged at the cathode. The cathode of the battery draws electrons from the anode of the electrolytic cell. The anions in the electrolyte are attracted to the positively charged anode, where they give up their electrons to become electrically neutral and are also discharged.

Hence, an electric current passes through the complete circuit.

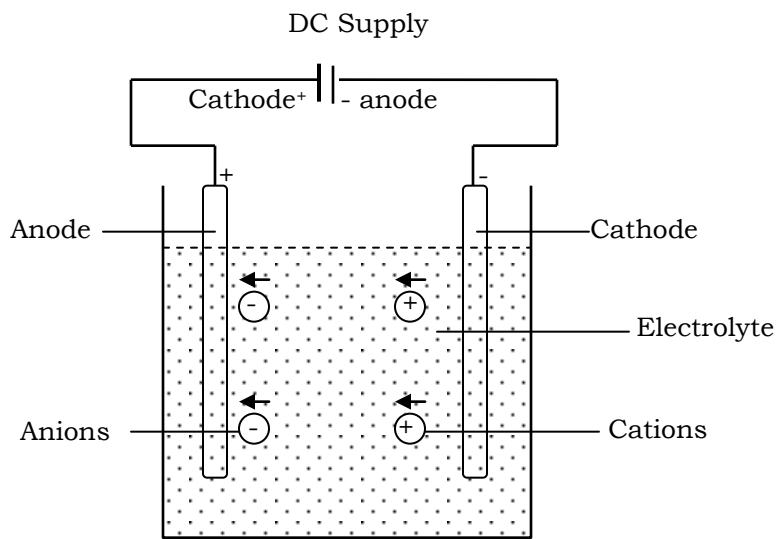


Fig. 1: *Mechanism of electrolysis*

The required product of electrolysis are in different physical states from the electrolyte and can be removed by some physical processes. For example, in the electrolysis of brine to produce hydrogen and chlorine, the products are gaseous. These gaseous products bubble from the electrolyte and are collected. In general, metals or hydrogen gas are discharged at the cathode while non-metals (except hydrogen) are discharged at the anode.

Electrolysis of Acidified Water

Acidified water is made by adding a few drops of tetraoxosulphate (vi) acid to water. Therefore, the electrolysis of acidified water is the same as the electrolysis of dilute tetraoxosulphate (vi) acid.

Acidified water contain the following:

	H_2SO_4	H_2O
Cations	$2\text{H}^+(\text{aq})$	$\text{H}^+(\text{aq})$
Anions	$\text{SO}_4^{2-}(\text{aq})$	$\text{OH}^-(\text{aq})$
Concentration of ions	High	Low
Position in electrochemical series	OH^- are higher in the series than SO_4^{2-}	

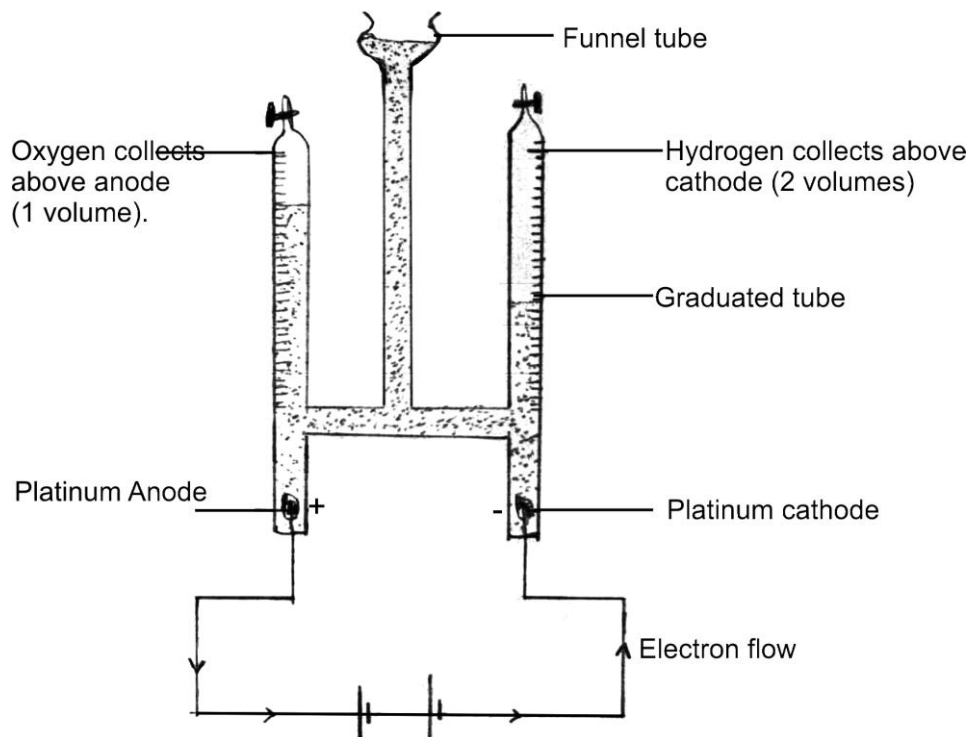
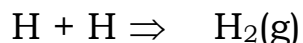
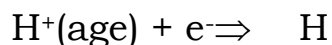
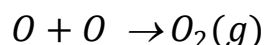
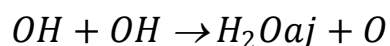
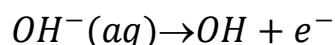


Fig. 2: Electrolysis of acidified water

At the Cathode: The H^+ ions migrate to the cathode where they acquire an electron each to form neutral hydrogen atoms. These atoms then combine in pairs to produce diatomic molecules of gaseous hydrogen.



At the anode: Both SO_4^{2-} and OH^- migrate to the anode, where OH^- are preferentially discharged. Each OH^- loses its electron to the anode to become a neutral $-OH$ group. Two $-OH$ groups react to form one molecule of water and one atom of oxygen. Gaseous oxygen is produced when the oxygen atoms combine in pairs to form diatomic molecules.



Summary

Cathodic Half-reaction: $4H^+(aq) + 4e^- \rightarrow 2H_2(g)$

Anodic half-reaction: $4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$

Overall reaction: $4H_2O(l) \rightarrow 4H^+(aq) + 4OH^-(aq)$
 $\rightarrow 2H_2(g) + O_2(g) + 2H_2O(l)$
 $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

The electrolysis of acidified water yields 2 volumes of hydrogen at the cathode and 1 volume of oxygen at the anode, i.e. the ratio of hydrogen to oxygen by volume is 2:1 respectively.

FARADAY'S LAW OF ELECTROLYSIS

Faraday's laws of electrolysis are quantitative relationships based on the electrochemical researches published by Michael Faraday in 1834.

Several versions of the laws can be found in textbooks and in scientific literature. The most common statements resemble the following:

Faraday's First Law of Electrolysis: The mass of a substance altered at an electrode during electrolysis is directly proportional to the quantity of electricity transferred at that electrode. Quantity of electricity refers to the quantity of electric charge, measured in coulomb.

$$M \propto Q \quad \dots\dots\dots (1)$$

$$M \propto It \quad \dots\dots\dots (2) \text{ where } Q = It$$

$$M = EIt \quad \dots\dots\dots (3)$$

E is a constant which is defined as the mass in grams of a substance liberated by the passage of 1 coulomb of electricity.

Faraday's Second Law of Electrolysis: When the same quantity of electricity is passed through different electrolytes, the relative number of moles of the elements discharged are inversely proportional to the charges on the ions of the elements.

The minimum quantity of electricity required to liberate one mole of singly-charged ions, such as hydrogen or chloride ions, is 96500 coulombs. This quantity of electricity is called the Faraday and is denoted by F. This follows that two faradays will be needed to discharge one mole of ions of a divalent or radical such as zinc or tetraoxosulphat (vi).

Faraday's two laws can be summarized as follows:

$$M = \left(\frac{Q}{F}\right) \left(\frac{M}{V}\right) \dots\dots\dots (4)$$

Where M is the mass of the substance liberated at an electrode in grams, Q is the total electric charge pass through the substance F = 96485Cmol⁻¹ is the Faraday Constant, M is the molar mass of the substance and finally v is the valency number of the ions of the substance.

For Faraday's First Law, M, F, and V are constants, so that the larger the value of Q the larger M will be. For Faraday's Second

Law, Q, F, and v are constants, such that the larger the value of the equivalent weight the larger M will be.

Equation (4) can also be written as:

$$M = \left(\frac{It}{F}\right) \left(\frac{M}{V}\right) \dots\dots\dots (5)$$

Example 1

A charge of $2 \times 10^{-4}F$ is passed through an electrolytic containing ferric ions (Fe^{+++}). Assuming that the only cathode reaction is $Fe^{+++} + 3e^{-} \rightarrow Fe$, What mass of iron will be deposited (Atomic weight of iron is 55.85).

$$M = ?$$

$$Q = 2 \times 10^{-4}F$$

$$M = 55.85$$

$$V = 3$$

Recall that:

$$\begin{aligned} M &= \left(\frac{Q}{F}\right) \left(\frac{M}{V}\right) = \frac{2 \times 10^{-4}F}{F} \times \frac{55.85}{3} \\ &= 2 \times 10^{-4} \times \frac{55.85}{3} \\ &= 3.72 \times 10^{-3}kg = \mathbf{3.72g} \end{aligned}$$

Example 2

Calculate the mass of silver deposited when a current of 2.6A is passed through a solution of a silver salt for 70minutes ($Ag = 108$; 1 Faraday = 96500C).

$$M = 108$$

$$F = 96400C$$

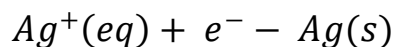
$$t = 70 \times 605 = 42005$$

$$I = 2.6A$$

$$M = ?$$

$$V = i$$

$$M = \left(\frac{Q}{F}\right)\left(\frac{M}{V}\right)$$



$$108_{(g)} \quad IF \quad 108_g$$

$$M = \left(\frac{It}{F}\right)\left(\frac{M}{V}\right) = \frac{2.6 \times 4200}{96500} \times \frac{108}{1}$$

$$= \underline{\underline{12.22g}}$$

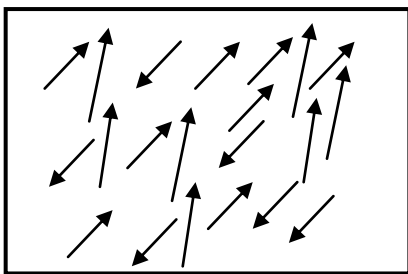
MAGNETISM

Magnets are objects that produce magnetic fields and attract metals like iron, nickel and cobalt. Magnetic field lines of force enter a magnet from the south pole and exit at the north pole. Permanent or hard magnets create their own magnetic fields all the time. While temporary or soft magnets produce magnetic fields while in the presence of a magnetic field and for a short while after exiting the field. Electro-magnets on the other hand, produce magnetic fields only when electric current travels through their coils.

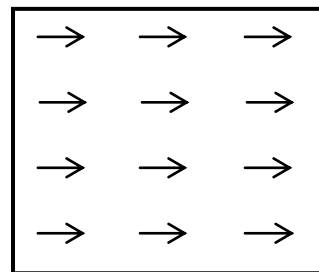
In the past magnets were usually made of metals or metallic alloys. But these days you have magnets made from other materials such as ceramic, Alnico (Aluminum, nickel, and cobalt) Neodymium, Samarium Cobalt and Polymers. What is responsible for magnetism in magnets? To find the answer, just Read on.

Magnetic Domains

Each electron in an atom acts like a tiny electromagnet. The magnetic fields of a group of atoms close to each other combine together to form a domain. Although magnetic domains can contain as much as 10^{20} electrons they are still usually very small ranging from around 10 to 1000 microns. Thus a small sample of iron can contain a huge number of magnetic domains. When a piece of metal is not magnetized, its magnetic domains point in random directions metal or a temporary magnet, the domains tend to align with the external field, and return to their random arrangements when the field is removed.



Domain of a piece of iron

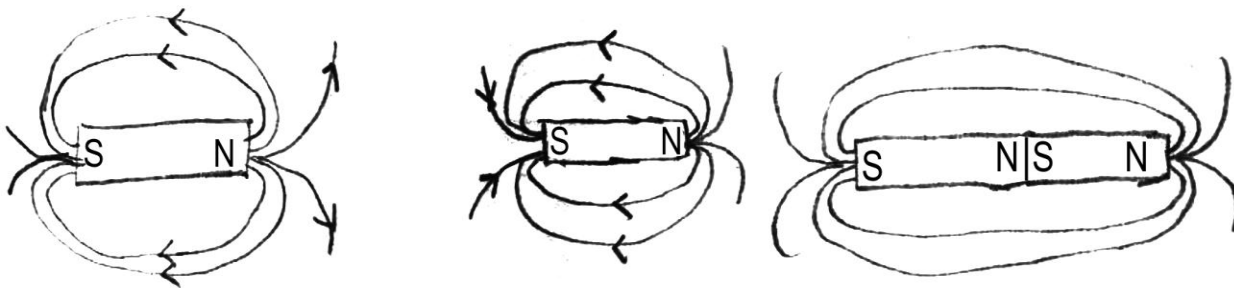


Domain of a magnet

Forces of Objects in Magnetic Fields

When a sample made of iron, cobalt, or nickel is placed in the magnetic field of a permanent magnet, the field lines become concentrated within the sample. Lines leaving the N – pole of the magnet enter one end of the sample, pass through it, and leave at the other end. Thus, the end of the sample closest to the magnet's N-pole becomes the sample's S-pole and the sample is attracted to the magnet.

Magnetic fields also exert forces on other magnets. The field produced by the N-pole of one magnet pushes out and naturally enters the south pole of another magnet, the attractive force then creates another larger magnet. Like poles repel each other because their lines of force are traveling in opposite directions, clashing with each other rather than moving together.



The figures above show to separate magnets be and after the are brought close to each other.

Magnetic Fields

The region round a magnet, where a magnetic force occurs, is called a magnetic field. The magnetic field lines are drawn by means of a compass needle placed near the magnet. The direction of the magnetic field at a given point is the direction of the north pole of the compass needle points. Magnetic field lines are drawn so that the tangent to a field line at any point gives the direction of the magnetic field at that point, and they are also drawn so that the number of lines per unit cross-sectional area is proportional to the magnitude of the magnetic field.

Experimentally it is found that when a charged particle (either alone or as part of a current) moves through a magnetic field, a force due to the field can exert a force on the particle. If we fire a charged particle through a point at which a magnetic field $\vec{B}$ is to be defined, using various directions and speeds for the particle and determining the force F_B that acts on the particle at that point. We find that when the particle's velocity is along a particular axis through the point, force F_B is zero. For all other directions of $\vec{V}$ the magnitude of F_B is zero. For all other directions of $\vec{V}$ the magnitude of F_B is always proportional to $V\sin\theta$. Where θ is the angle between the zero – force axis and the direction of $\vec{V}$. Hence, the direction of F_B is always perpendicular to the direction of $\vec{V}$.

$$\therefore \quad \vec{F}_B = q \vec{V} \times \vec{B} \quad \dots\dots\dots (1)$$

The force $\vec{F}_B$ a particle is equal to the charge q times the cross product of its velocity and the field $\vec{B}$.

$$\vec{F}_B = q/vB\sin\theta \dots\dots\dots (2)$$

The SI unit for $\vec{B}$ that follows equation (2) is the newton per coulomb-meter per second. For convenience, this is called tesla(T) after Nikola Tesla a Serbian – American inventor. An earlier (non-SI) unit for $\vec{B}$, still in common use, is the gauss (G).

$$1\text{tesla} = 10^4\text{gauss}.$$

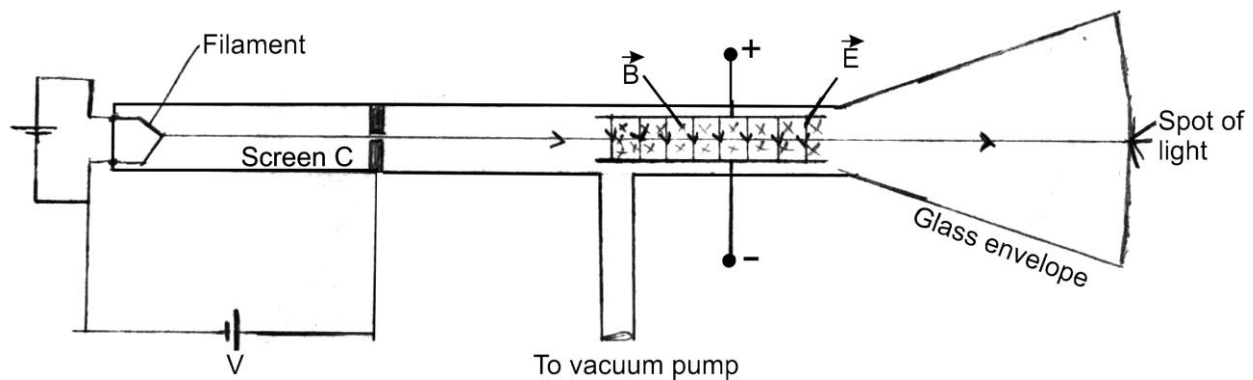
$$1\text{tesla} = \text{IT} = \frac{\text{newton}}{(\text{coulomb})(\text{meter /second})} = 1 \frac{\text{N.s}}{\text{C.m}}$$

When both magnetic field $\vec{B}$ and an electric field E are present in a region, the force on a test charge is the vector sum of the electric and magnetic forces: $F_{\text{tot}} = q [E + (v \times B)]$. This relation is known as the Lorentz equation.

$$F_{\text{tot}} = q [E + (v \times B)] \dots\dots\dots (3)$$

Crossed Fields

Both electrical field $\vec{E}$ and magnetic field $\vec{B}$ can produce a force on a charged particle. When the two fields are perpendicular to each other, they are said to be crossed fields. J J Thompson in 1897 used crossed fields in his experiments that led to the discovery of the electron. A modern version of Thompson's apparatus is described below.



Charged particles (which are now known as electrons) are emitted by a hot filament at the end of the tube and accelerated by an applied potential difference V . After they pass through a slit in the screen C, they form a narrow beam. They then pass through a region of crossed $\vec{E}$ and $\vec{B}$ fields, and head toward the fluorescent screen S where they produce a spot of light. By controlling the magnitudes and directions of the fields, Thompson could control where the spot of light appeared on the screen. For the arrangement on the figure above, electrons are forced up by the electric field $\vec{E}$ and down the page by the magnetic field $\vec{B}$.

From our knowledge of electric fields we know that when a charge particle(q) is moving in an electric field $\vec{E}$ with a velocity V and mass M in a plate of length L the deflection of the particle y is given by:

$$y = \frac{2EL^2}{2mv^2} \dots\dots\dots (4)$$

$$E = \frac{2mv^2y}{qL^2} \dots\dots\dots (5)$$

and that: $\vec{E} = \frac{\vec{F}_c}{q} \dots\dots\dots (6)$

$$\vec{F}_E = /q/E \dots\dots\dots (7)$$

Where the two fields are adjusted so that the two deflecting forces cancel, we have that:

$$/q/E = /q/VB\sin 90^\circ$$

$$/q/E = /q/VB$$

$$V = \frac{E}{B} \dots\dots\dots (8)$$

If we put equation 8 into equation 4 we have that:

$$y = \frac{qEL^2}{2m\left(\frac{E^2}{B^2}\right)} = \frac{qEB^2L^2}{2mE^2}$$

$$y = \frac{qB^2L^2}{2mE}$$

$$2mEy = 2B^2L^2$$

$$m = \frac{qB^2L^2}{2Ey}$$

Dividing both sides by the charge q we get.

$$\frac{m}{q} = \frac{B^2L^2}{2Ey} \dots\dots\dots (9)$$

Thus crossed field allow use to measure the ratio of m/q of particles moving through Thompson's apparatus.

Example

A He^{2+} ion travels at right angles to a magnetic field of 0.80T with a velocity of 10^5 m/s . Find the magnitude of the magnetic force on the ion.

$$\theta = 90^\circ \quad V = 10^5 \text{ m/s}$$

$$\vec{B} = 0.80 \text{ T} \quad F = ?$$

Force is given by:

$$F = qvB \sin \theta$$

$$F = 2(1.60 \times 10^{19}) \times 10^5 \text{ m/s} \times 0.80 \text{ T} \sin 90^\circ$$

$$\mathbf{F = 2.56 \times 10^{14} N}$$

HALL'S EFFECT

We have earlier proved that a beam of electrons in a vacuum can be deflected by a magnetic field. However, can the drifting conduction electrons in a copper wire also be deflected by a magnetic field? This is what Edwin H. Hall in 1879 showed.

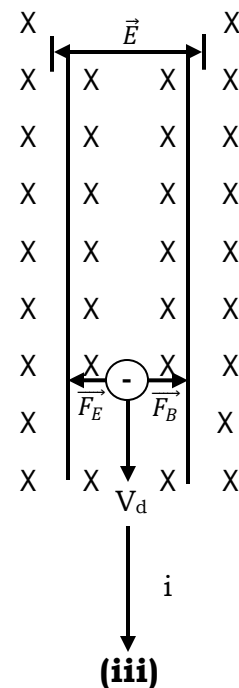
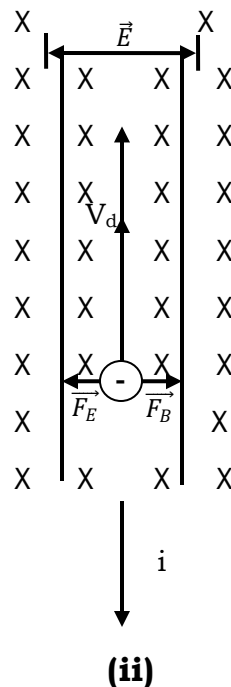
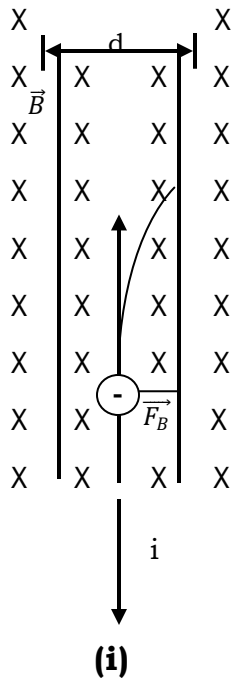


Fig (i) shows a copper strip carrying a current i whose direction is from the top to the bottom of the figure. If any external magnetic field $\vec{B}$, pointing into the plane of the figure, we see that the deflecting force $\vec{F}_B$ will act on each electron drifting it toward the right edge of the strip. As time passes electron pill at the right leaving the left negative. The separation of positive and negative charges produce an electric field $\vec{E}$ with the strip, pointing left. As time continues to pass by equilibrium develops and the force due to $\vec{B}$ and that due to $\vec{E}$ balances each other allowing the electrons to drift out without further increase in $\vec{E}$.

By connecting a voltmeter across the width of the strip, we can measure the Hall potential difference between the two edges of the trip.

Suppose V_H is the magnitude of the Hall voltage and d is the width of the strip. Then the electric field strength E set up across the strip is $E = V_H/d$ (10)

Now when the electric magnetic forces are in equilibrium,

$$eE = eV_d B \quad \text{.....} \quad (11)$$

However drift velocity is $V_d = \frac{J}{ne} = \frac{i}{neA}$ (12)

Where $J = 1/A$ is the current density of the strip, A is the cross-sectional area of the strip and n is the number density of charge carriers.

If we put equation (12) and equation (10) into equation (11) we get.

$$E = V_d B$$

$$E = \frac{iB}{neA}$$

$$\frac{V_H}{d} = \frac{iB}{neA}$$

$$V_H = \frac{iB}{neA}$$

$$V_H = \frac{iB}{ne} \left(\frac{A}{d}\right)^{-1}$$

In which $1(=A/d)$ is the thickness of the strip.

$$\therefore V_H = \frac{iBL^{-1}}{ne}$$

$$V_H = \frac{Bi}{nel} \dots\dots\dots (13) \text{ Halls potential difference}$$

Also,

$$n = \frac{Bi}{V_H el} \dots\dots\dots (14) \text{ number of density}$$

change carriers.

LECTURE NOTE ON

- **TRANSFORMER**
- **A.C. CIRCUIT**
- **MAGNETIC PROPERTIES OF MATTER**
- **MAGNETIC FORCE ON A CURRENT CARRYING CONDUCTOR**

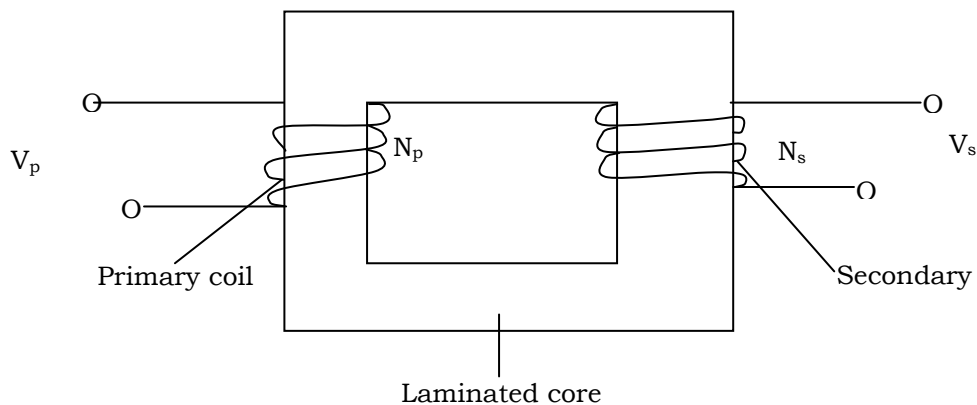
By:

OMOJOLA

TRANSFORMER

This is an electrical device used to step-up or step-down voltage in an a.c. circuit.

It consists of a primary and a secondary coil wound on the same iron core. See fig. 1.0 below.



The basic principle of operation of a transformer is based on Faraday's law of electromagnetic induction.

The law which states that whenever there is a change in the magnetic flux linking a circuit, an e.m.f. is directly proportional to the time rate of change of magnetic flux linking the circuit.

$$E \propto \frac{d\phi_B}{dt} \Rightarrow E = \frac{Nd\phi_B}{dt}$$

Working

If the primary coil is connected to a source, it creates a varying magnetic flux in the core. This changing magnetic flux is connected

to the secondary coil and give rise to induced e.m.f. in the coil that is proportional to the number of turns in the coil.

$$\text{i.e. } \frac{V_s}{V_p} = \frac{N_s}{N_p} \dots\dots\dots (i)$$

Where: V_s = Induced e.m.f. or voltage in the secondary coil

V_p = Induced e.m.f. or voltage in the primary coil

N_s = Number of turns in secondary coil

N_p = Number of turns in primary coil

Ideal Transformer

An ideal transformer is one with no losses of any kind. i.e. a transformer with 100 percent efficiency.

Power in primary = power in secondary or

Input power = output power

$$I_p V_p = I_s V_s \dots\dots\dots (ii)$$

Where I_p and I_s are primary and secondary current

$$\text{Hence: } \frac{I_p}{I_s} = \frac{V_s}{V_p} \dots\dots\dots (iii)$$

But from equation (i) $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

$$\Rightarrow \frac{I_p}{I_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Where $\frac{N_s}{N_p}$ is called the turn ratio.

By choosing the appropriate turns ratio $\frac{N_s}{N_p}$, we may obtain any desired secondary voltage from a given primary voltage.

If $N_s > N_p$, then $V_s > V_p$ and we have a step-up transformer.

If $N_s < N_p$, then $V_s < V_p$ and we have a step-down transformer.

At a power generating station, step-up transformer are used; the primary is connected to the power source and the secondary is connected to the transmission lines giving the desired high voltage for transmission. Near the consumer, a step-down transformer lower the voltage to a value suitable for use in home or industry.

ENERGY LOSSES IN A TRANSFORMER AND HOW TO MINIMIZE IT

Copper or heat losses – energy is lost in form of heat. I^2R in the coil	Using a low resistant coil or thick wire.
Eddy current losses – magnetic flux induces current in the core, referred to as eddy currents, thereby heating up the core	Is minimized by increasing the resistance of the core by the lamination of the iron core.
Hysteresis losses – This happens in the transformer core since the core is taken through a cycle. The loss is dependent on the area of the hysteresis loop.	By use of core made of soft magnetic core.
Magnetic leakage – some magnetic flux does not pass through the iron core	By special forms of coil winding.

Types of Transformer

Based on construction we have:

- (i) Core – type transformer
- (ii) Shell – type transformer
- (iii) Spiral – type

Efficiency of a Transformer

$$\text{Efficiency} = \frac{\text{Power in secondary}}{\text{power in primary}} \times 100\%$$

$$E = \frac{I_s V_s}{I_p V_p} \times 100\%$$

Example:

1. What is the turn ratio required in a transformer to step down 250V to 50V. If the secondary current is 2A and resistance in the primary is 10kn. Calculate the primary current.

Solution

$$\frac{\text{Primary Voltage } V_p}{\text{Secondary Voltage } V_s} = \frac{\text{Primary turn } N_p}{\text{Secondary turn } N_s}$$

$$\frac{250}{50} = \frac{N_p}{N_s} = 5 \text{ turn ratio}$$

Since $I_s = 2A$

From the relation $\frac{I_p}{I_s} = \frac{N_s}{N_p}$

$$\begin{aligned} \therefore I_p &= I_s \times \frac{N_s}{N_p} \\ &= 2 \times \frac{1}{5} = 0.4A \end{aligned}$$

$\therefore$ The primary current $I_p = 0.4A$

2. A step-up transformer is used on a 120V line to furnish 1800V. The primary has 100 turns. How many turns are on the secondary?

Solution

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} \Rightarrow \frac{120}{1800} = \frac{100}{N_s}$$
$$\Rightarrow N_s = 1500 \text{ turns.}$$

3. A transformer used on a 120V line deliver 2.0A at 900V. What current is drawn from the line? Assume 100% efficiency.

Solution

Power in primary = Power in Secondary

$$I_p V_p = I_s V_s$$

$$I_p \times 120 = 2 \times 900$$

$$\therefore I_p = 15A$$

Exercise

1. A step-up transformer operates on a 110V line and supplies a load with 5.0A. The ratio of the primary and secondary windings is 1:25. Determine :

- (i) The secondary voltage
- (ii) The primary current and
- (iii) The power output. Assume a resistive load and 100% efficiency.

2. An ideal step-down transformer has a turn ratio of 20. The r.m.s. current and voltage in the primary are 20A and 250V. calculate:
 - (i) The maximum current and voltage in the primary
 - (ii) The r.m.s. current and voltage in the secondary.
3. A step-down transformer is used on a 1650V line to deliver 45A at 110V – What current is drawn from the line? Assume 100% efficiency.

A.C. CIRCUITS

An alternating current is a current that varies sinusoidally.

The current, emf or voltage of an a.c. source is represented as:

$$\begin{aligned}
 I &= I_o \sin \omega t \\
 &= I_o \sin \bar{\lambda} f t \quad \text{where } \omega = 2\pi f \\
 V &= V_o \sin \omega t \\
 &= V_o \sin \bar{\lambda} f t
 \end{aligned}$$

Where:

$$\begin{aligned}
 I &= \text{Instantaneous current} \\
 I_o &= \text{Peak or maximum current} \\
 V_o &= \text{Instantaneous voltage} \\
 V_o &= \text{Peak or maximum voltage} \\
 \omega &= \text{angular velocity} \\
 \omega t &= \text{Phase angle of the current.}
 \end{aligned}$$

The output of an a.c. source is shown in fig. 2.0 below.

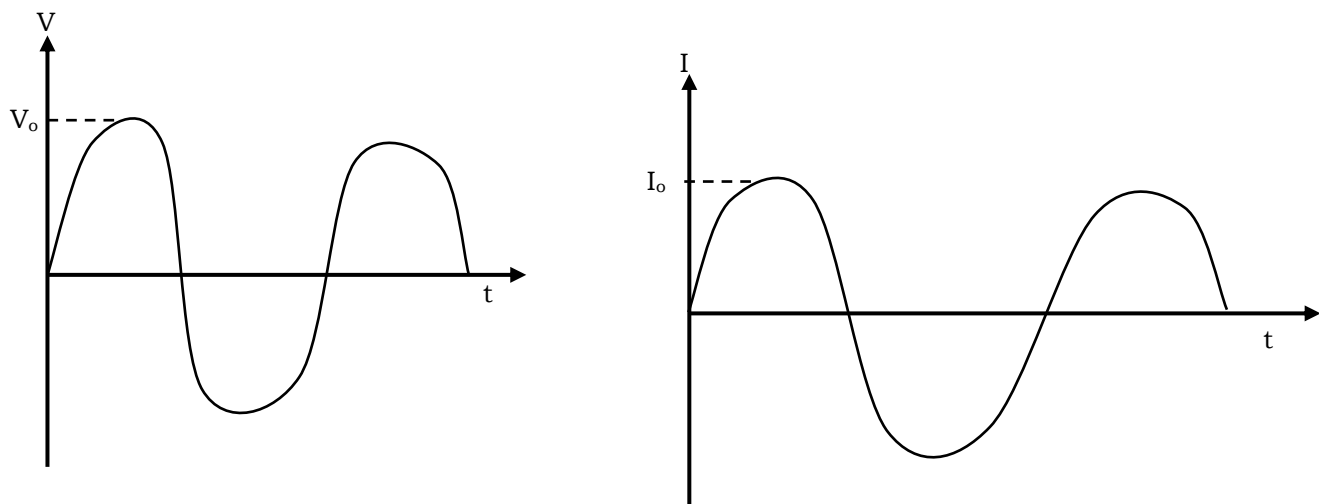


Fig. 2.0: *An a.c. output*

THE PEAK (AMPLITUDE) AND ROOT MEAN SQUARE (R.M.S) VALUES

The peak value is the maximum current recorded during an a.c. cycle.

R.m.s value of an alternating current is the direct current which gives the same heating rate as a given alternating current. It is also called effective value of the current. It is given by:

$$I_{r.m.s} = \frac{I_o}{\sqrt{2}} = 0.707I_o$$

$$V_{r.m.s} = \frac{V_o}{\sqrt{2}} = 0.707V_o$$

RESISTOR IN AN A.C. CIRCUIT

Fig. 2.1(a) show a simple a.c. circuit where an a.c. voltage is connected to a resistor. Fig. 2.1(b) show the wave form and fig. 2.1(c) the phase diagram

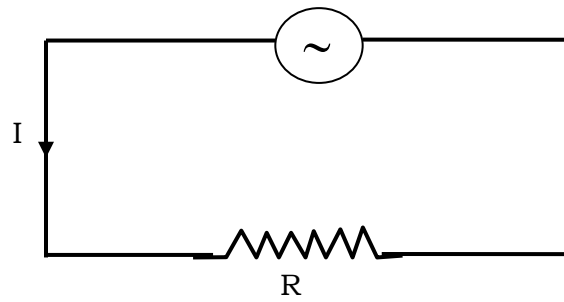


Fig. 2.1(a)

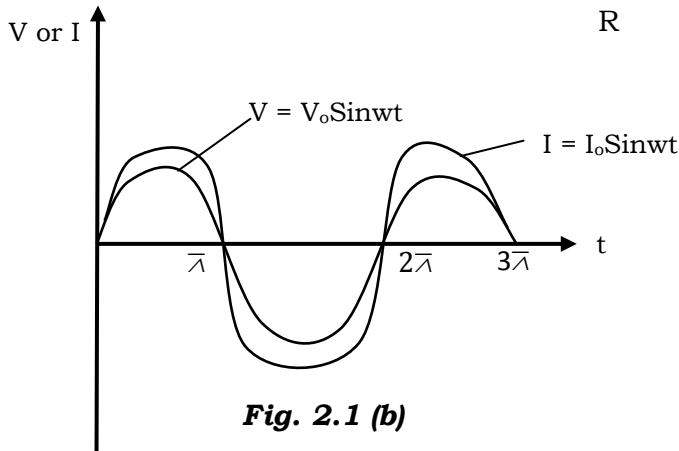
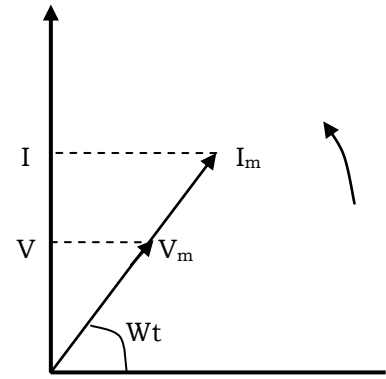


Fig. 2.1 (b)



Phasor diagram

When an a.c. voltage is applied to a pure resistor, the voltage across the resistance and current through it attain their maximum values at the same instant and their zero values at the same instant. The voltage and current are said to be in phase. i.e. from ohm's law we have

$$V = IR$$

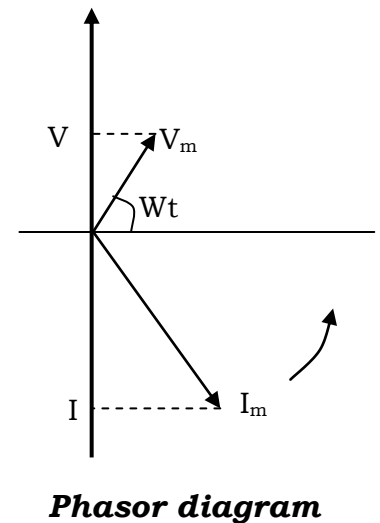
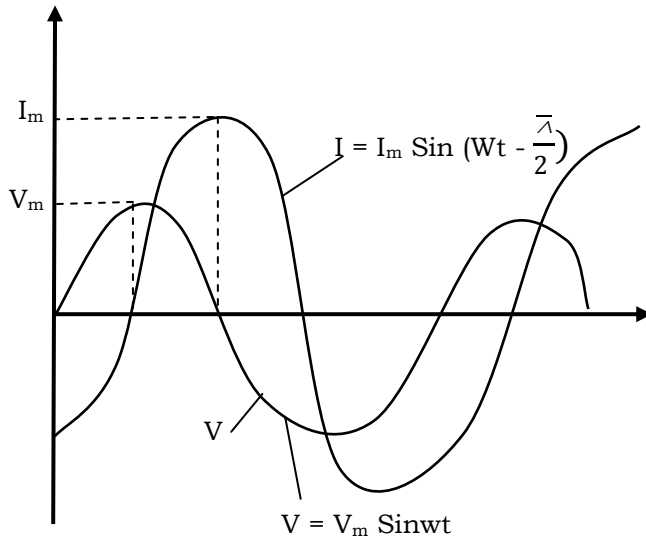
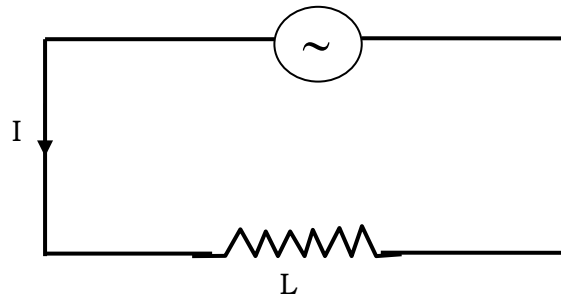
$$\Rightarrow I = \frac{V}{R}$$

Putting $V = V_o \sin \omega t$]

$$\Rightarrow I = \frac{V}{R} = \frac{V_o \sin \omega t}{R} = I_o \sin \omega t$$

$$\therefore I_{r.m.s} = \frac{V_{r.m.s.}}{R}$$

INDUCTOR IN AN A.C. CIRCUIT



When an a.c. voltage is applied to a pure inductance, the voltage across the inductance reach its maximum value $\frac{1}{4}$ cycle ahead of the current i.e. when the current is zero, the back emf of the inductance causes the current through the inductance to lag behind the voltage by $\frac{1}{4}$ cycle (or $\frac{\pi}{2}$), and the two are 90° out of phase.

If $V = V_m \sin \omega t$ the current will be

$$I = I_m \sin (\omega t - \frac{\pi}{2}),$$

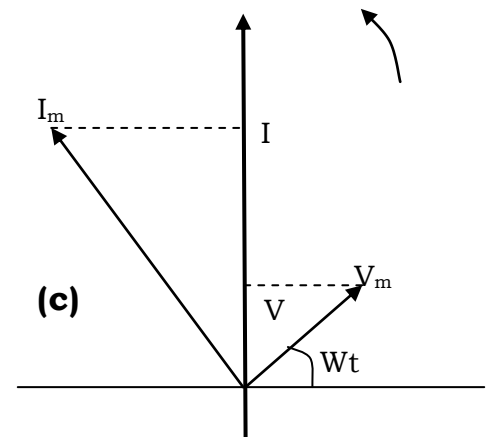
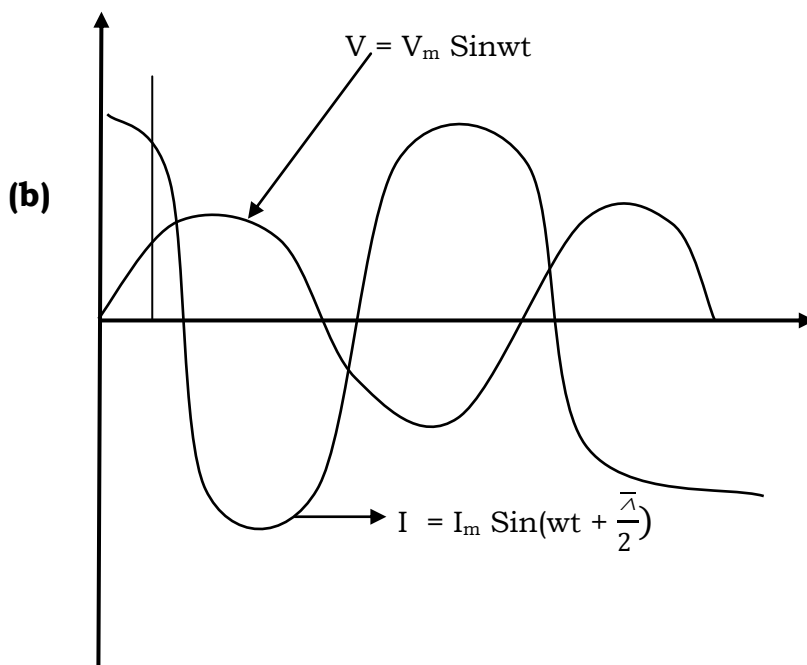
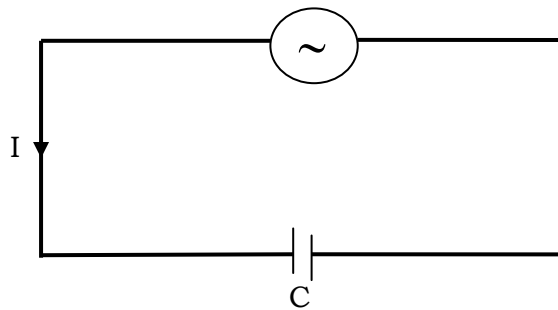
The inductor opposes the flow of current. This opposition offered to the flow of a.c. by the inductor is known as inductive reactance X_v .

This is given by the relation:

$$X_v = 2\pi fL \quad V = IX_v$$

CAPCITOR IN AN A.C. CIRCUIT

A capacitor with capacitance C is connected to the a.c. source as shown in fig. 2.3



Graphical rep.

When an a.c. voltage is applied to a pure capacitor, the voltage across it lags 90° behind the current flowing through it. Current must flow before the voltage across (and charge on) the capacitor can build up.

If $V = V_o \sin \omega t$ the current will be

$$I = I_o \sin \left(\omega t - \frac{\pi}{2} \right),$$

Capacitor opposes the flow of current. This opposition is called capacitive reactance.

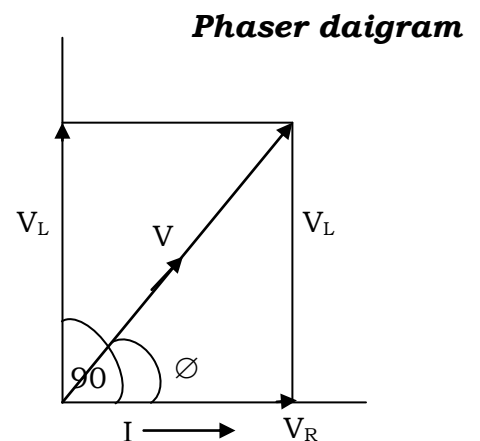
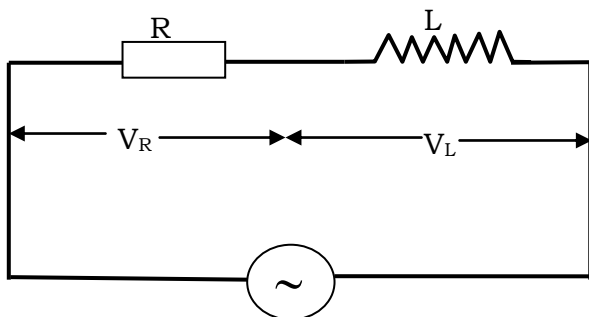
This is given by:

$$X_c = \frac{1}{2\pi f C} \quad V = IX_c$$

Series Circuit

L and R in series

Consider an inductor L in series with resistance R, with an alternating voltage



In R. L. circuit, the sum of the respective voltages V_R and V_L across R and L is equal to V. But the voltage V_L leads by 90° on the current I, and the voltage is in phase with I. thus the voltages can be drawn to scale as shown in fig. 2.4(b) and hence by Pythagoras theorem

$$V^2 = V_L^2 + V_R^2$$

$$\text{But } V_L = IX_L$$

$$\Rightarrow V^2 = I^2 X_L^2 + I^2 X_R^2$$

$$V^2 = I^2 (X_L^2 + X_R^2)$$

$$\Rightarrow I^2 = \frac{V^2}{X_L^2 + X_R^2} \therefore I = \frac{V}{\sqrt{X_L^2 + X_R^2}}$$

From fig. 2.4(b), the current lags behind the applied voltage V by an angle given by

$$\tan\theta = \frac{V_L}{V_R} = \frac{IX_L}{IR} = \frac{X_L}{R} \quad \dots\dots\dots (b)$$

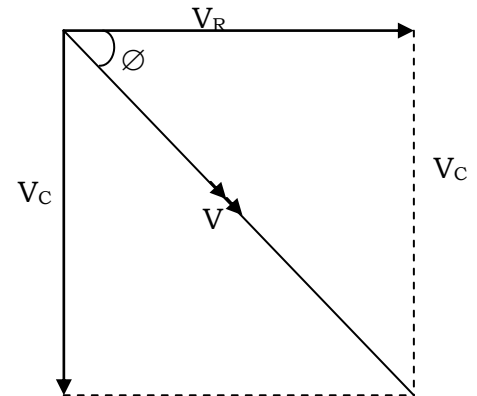
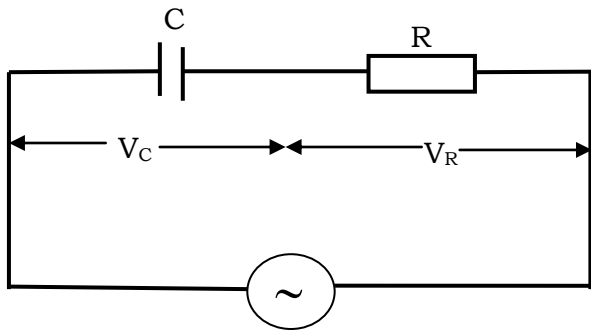
$$\therefore \theta = \tan^{-1}\left(\frac{X_L}{R}\right)$$

From equation (a) opposition Z to the flow of a.c. is given by

$$Z = \frac{V}{I} = \sqrt{X_L^2 + R^2}$$

Where Z = Impedance

C and R in series



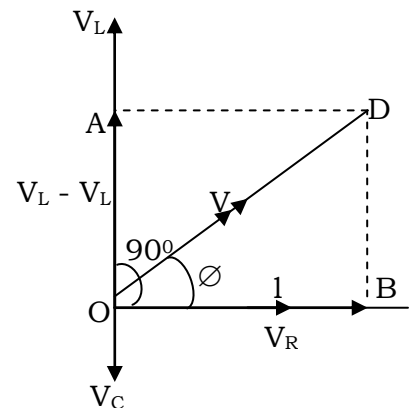
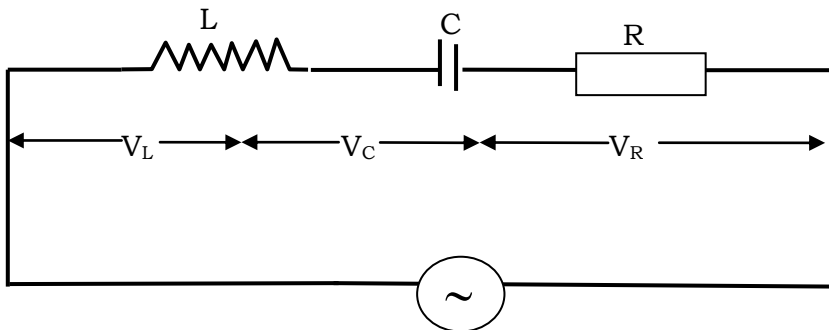
Apply similar analysis to that of L and R in series and find the impedance of a capacitor C and resistance R in series and the angle θ i.e.

$$Z = \frac{V}{I} = \sqrt{X_C^2 + R^2}$$

$$\tan \theta = \frac{X_C}{R} \Rightarrow \theta = \tan^{-1} \left(\frac{X_C}{R} \right)$$

L.C.R. in series

Consider a circuit containing an inductance (L) capacitance (C) and Resistance (R) in series



The phasor diagram has V_L leading by 90° on V_R V_C lagging by 90° on V_R with the current I in phase with V_R fig 2.6(b). If V_L is greater than V_C their resultant is $(V_L - V_C)$ in the direction of V_L as shown.

Consider triangle ODB

From Pythagorean theorem.

$$V^2 = (V_L - V_C)^2 + V_R^2$$

$$\text{But } V_L = IX_L, V_C = IX_C \quad V_R = IR$$

$$V^2 = (IX_L - IX_C)^2 + I^2 R^2$$

$$V^2 = I^2 [(X_L - X_C)^2 + R^2]$$

$$\Rightarrow I = \frac{V}{\sqrt{(X_L - X_C)^2 + R^2}} \quad \text{or} \quad I = \frac{V}{\sqrt{X^2 + R^2}}$$

$$\text{where } X = X_L - X_C$$

$$\therefore Z = \frac{V}{I} = \sqrt{(X_L - X_C)^2 + R^2}$$

Also fig. 2.6(b) I lags on V by angle ϕ given by

$$\tan \phi = \frac{DB}{OB} = \frac{V_L - V_C}{V_R} = \frac{IX_L - IX_C}{IR} = \frac{X_L - X_C}{R}$$

NB: Reactance X is the opposition offered to the flow of alternating current either by the capacitor or the inductance or both

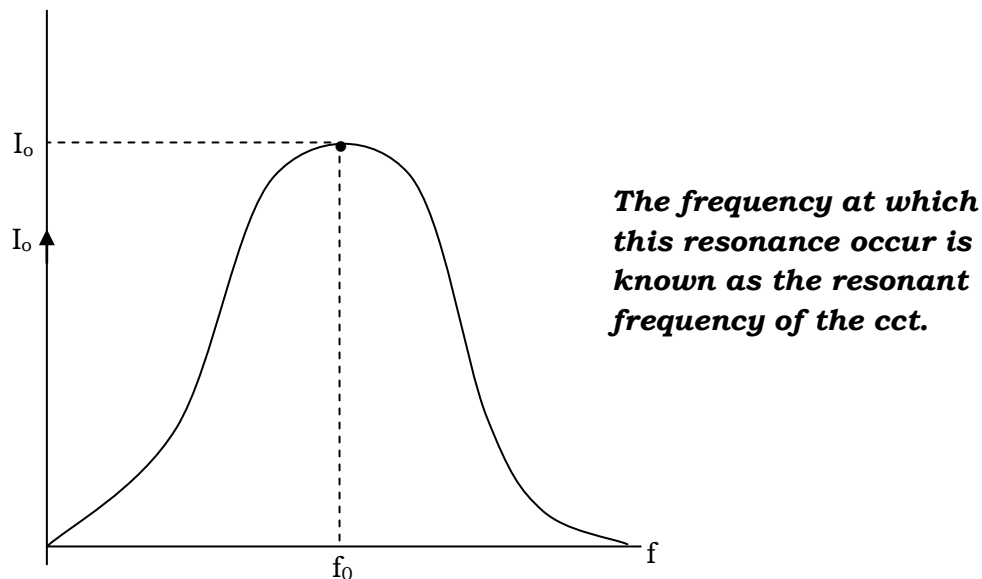
$$X = X_L - X_C$$

Impedance Z of the a.c. circuit is the total opposition offered to the flow of the a.c. by the resistor and either inductance or capacitor or both. It is measured in ohm's.

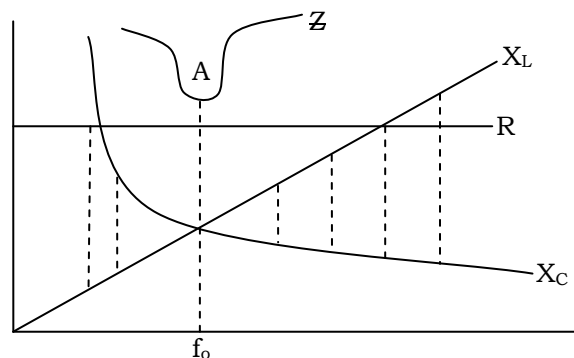
RESONANCE IN THE RLC SERIES CIRCUIT

Resonance is said to occur in an a.c. series circuit when the maximum current is obtained from such a circuit.

Consider the graph of variation of current I and frequency f .



Consider the graph of variation of Z with f



NB $X_L = 2\pi fL \Rightarrow X_L \propto f$

$$X_C = \frac{1}{2\pi fC} \Rightarrow X_C \propto \frac{1}{f}$$

Variation of X_L with frequency is a straight line passing through the origin. Variation of X_C with frequency is a curve approaching the two axis. R is independent of frequency and so it is represented by a line parallel to the frequency axis.

The diff ($X_L - X_C$) is represented by the broken lines. It can be seen that $X_L - X_C$ decrease to zero for a particular frequency f_0 and thereafter increases again so from $Z = \sqrt{(X_L - X_C)^2 + R^2}$, the impedance diminishes to A and then increases as the frequency f is varied.

The magnitude of f_0 is given by

$$X_L - X_C = 0 \Rightarrow X_L = X_C$$

$$2\pi f_0 L = \frac{1}{2\pi f_0 C}$$

$$4\pi^2 LC f_0^2 = 1$$

$$f_0^2 = \frac{1}{4\pi^2 LC} \Rightarrow f_0 = \frac{1}{\sqrt{2\pi LC}}$$

POWER IN AN A.C. CCT

In an a.c. circuit, the product of not mean square value of the current and the voltage does not give true power in Walt.

$P = I_{r.m.s.} V_{rms}$ (VA) i.e. is a reactive power measured as KVA.

The true power (W) in an a.c. circuit is given as:

$$P = IV \cos\phi$$

where $\cos\phi$ is called the power factor

$$\cos\phi = \frac{R}{Z}$$

$$\begin{aligned} \text{average power in a.c. } I_{r.m.s}^2 R &= I_{rms} V_{rms} \cos\phi \\ &= \frac{I_o V_o \cos\phi}{2} \end{aligned}$$

Examples

A voltage $V = (60.0V) \sin 120\pi t$ is applied across to 20.0Ω resistor.

What will an a.c. ammeter in series with the resistor read?

Solution

The r.m.s voltage across the resistor is

$$V_{r.m.s.} = \frac{V_o}{\sqrt{2}} = 0.707V_o.$$

$$V = 0.707 \times 60 = 42.4V$$

$$\text{From } V = IR \Rightarrow I = \frac{42.4}{20} = 2.12A$$

(2) A 120V a.c. voltage source is connected across a 2.0MF capacitor. Find (i) the current to the capacitor if the frequency of the source is 60HZ. (ii) What is the power loss in the capacitor.

Solution

From the relation.

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.142 \times 60 \times 2 \times 10^{-6}} = 1330\Omega$$

$$\text{Recall } V = IX_C \Rightarrow I = \frac{120}{1330} = 0.09A$$

$$\text{(ii) Since } \cos \phi = \frac{R}{Z} \text{ and } R = 0$$

$$\text{Power loss} = IV \cos \phi$$

$$= IV \cos 90 = 0$$

3. A 110V a.c. voltage source is connected across a pure 0.55H inductor. Find (i) The current through the inductor if the frequency of the source is 50Hz. (ii) What is the power loss in the inductor?

Solution:

$$X_L = 2\pi fL = 2 \times 3.142 \times 50 \times 0.55$$

$$= 172.81\Omega \simeq 173\Omega$$

then:

$$I = \frac{V}{X_L} = \frac{110}{173} = 0.635$$

$$\therefore \quad \mathbf{I = 0.635A}$$

4. A coil having inductance 0.18H and resistance of 14Ω is connected across a 120V, 30Hz line compute (a) the current in the

oil (b) the phase angle between the current and the supply voltage
 (c) the power factor and (d) the power loss in the coil

Solution

$$(a) X_L = 2\pi fL = 2 \times 3.142 \times 30 \times 0.18 = 33.9\Omega \approx 34\Omega$$

$$\text{and } Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{14^2 + (34 - 0)^2} = \sqrt{1352}$$

$$\therefore Z = 36.77\Omega$$

$$\text{So } I = \frac{V}{Z} = \frac{120}{36.77} = 3.26$$

$$\therefore I = 3.3A$$

$$(b) \tan \phi = \frac{X_L - X_C}{R} = \frac{34 - 0}{14} = 2.4$$

$$\phi = 67.6^\circ \approx \mathbf{68^\circ}$$

The voltage lead the current by 68°

$$(c) \text{ Power factor} = \cos \phi = \cos 68 = 0.37$$

$$(d) \text{ Power loss} = IV \cos \phi = 120 \times 3.3 \times 0.38$$

$$= \mathbf{0.15KW}$$

5. What will be resonant frequency of a series circuit containing an inductor of $200\mu H$ and a capacitor of $200pF$

Solution

$$f_o = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{6.28 \sqrt{200 \times 10^{-6} \times 200 \times 10^{-12}}}$$

$$= \frac{1}{6.28\sqrt{4 \times 10^{-14}}} = 796178.34$$

$$\therefore f_o = \mathbf{796.178KW}$$

Exercise

- (1) A series circuit consisting of a 100Ω non inductive resistor, a coil with a $0.20H$ inductance and negligible resistance, and a $20\mu F$ capacitor is connected across a $120V$ $60Hz$ power source. Find (a) the current (b) the power loss (c) the phase angle between the current and the source voltage and (d) the voltmeter readings across the three elements.
- (2) A current of $30mA$ is supplied to a $4.0\mu F$ capacitor connected across an alternating current line having a frequency of $500Hz$. Compute the reactance of the capacitor and the voltage across the capacitor.
- (3) An a.c. current in a 10Ω resistance produces thermal energy at the rate of $360W$. Determine the effective values of the current and voltage.

- (4) What is reactance and impedance of a 60Hz a.c. circuit containing an inductor of 2.5H, a resistor of 200Ω and a capacitor of $10\mu\text{F}$ connected in series.
- (5) What is the inductive reactance of the coil of 2H as a 60Hz alternating current flows through it?
- (6) Calculate the resonant frequency of a circuit of negligible resistance containing an inductance of 80mH and a capacitor of $600\mu\text{F}$.
- (7) A resistance of 20Ω , an inductance of 0.2H and a capacitance of $100\mu\text{F}$ are connected in series across 220V, 50Hz main. Determine the following:
- (a) Impedance, (b) current in the circuit.
 - (b) Voltage across R, L and C
 - (c) Power in volt ampere and in watts.
 - (d) The power factor
 - (e) Phase angle.

MAGNETIC PROPERTIES OF MATTER

Magnetism is a property of matter and it occurs in different forms and degree in various conductors and insulators. For example, at low temperature, metallic system exhibit either superconducting or magnetic order. The degree of magnetism of a substance is due to the intrinsic magnetic dipole moments of its electrons. The degree of magnetism is also called magnetization and it is defined as the net magnetic dipole moment of the substance per unit volume.

Faraday was the first to start classifying substance according to their magnetic properties. Faraday classified them as either diamagnetic or paramagnetic and he based his classification on the force exerted on the materials when placed in an inhomogeneous magnetic field.

Diamagnetic substance have a negative magnetic susceptibility, (i.e. they are materials in which the magnetization and magnetic field are opposite). The electrons in the atom of diamagnetic materials are all paired and there is no intrinsic Magnetic moment. When a material is placed into a magnetic field, its atoms acquire an induced magnetic moment pointing in a direction opposite to that of the external field and the material becomes magnetic. The

diamagnetic field produced by the material opposes the external field, although this diamagnetic field is very weak (except in superconductors).

NB: If the atoms of a material have no magnetic moment of their own, then diamagnetism is the only magnetic property of the material and the material is called diamagnetic. Copper exhibits such diamagnetism.

Paramagnetic substance has a weak positive magnetic susceptibility resulting from the presence of atoms (or ions) that have permanent magnetic moments. These moments interact only weakly with each other and are randomly oriented in the absence of an external magnetic field. When a paramagnetic substance is placed in an external magnetic field, its atomic moments tend to line up with the field. However, this alignment process must compete with thermal motion, which tends to randomize the magnetic moment orientation. If only a relatively small fraction of the atoms are aligned with the field, then the magnetization obeys Curie's law.

Curie's Law states that the magnetization of a paramagnetic substance is proportionate to the applied magnetic field and inversely proportional to the absolute temp.

$$M = C \frac{B_0}{T} \dots\dots (1) \text{ Where } C = \text{Curie's constant}$$

B_0 = Magnetic field

T = absolute temp

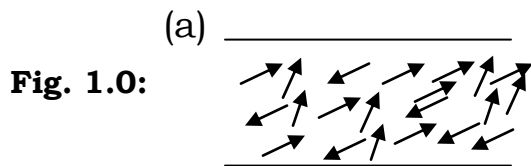
M = Magnetization

The law shows that when $B_0 = 0$, the magnetization is Zero, corresponding to a random orientation of magnetic moments. As the ratio of magnetic field to temperature becomes great, the magnetization approaches its saturation value corresponding to a complete alignment of its moments and equation is no longer valid.

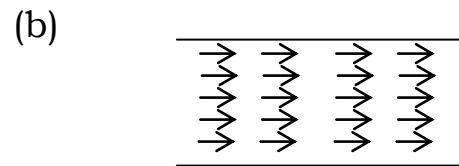
When the temp. of a ferromagnetic substance reaches or exceeds critical temp. called the Curie temperature, the substance loses its residual magnetization and becomes paramagnetic. Below the Curie temp., the magnetic moments are aligned and the substance is ferromagnetic. Above the Curie temp., the thermal agitation is great enough to cause a random orientation of the moments, and the substance becomes paramagnetic.

Ferromagnetic substance have the highest magnetic susceptibility. These substance contain atomic magnetic moment that tend to align parallel to each other even in a weak external magnetic field. Once the moments are aligned the substance remains magnetized after the external field is removed. This permanent alignment is due to a strong coupling between neighbouring moments, a coupling that can be understood only in quantum – mechanical terms.

All ferromagnetic materials are made up of microscopic regions called domains, region within which all magnetic moments are aligned. These domains have volumes of about 10^{-12} to 10^{-8} and contain 10^{17} to 10^{21} atoms. The boundance between the various domains having different orientations are called domain walls. In an unmagnetized sample, the domains are randomly oriented so that the net magnetic moment is zero as shown in fig. 1.0(a). When the sample is placed in an external magnetic field, the magnetic moments of the atoms tends to align with the field, which results in a magnetized sample.



Random orientation of a tomic magnetic moments in an unmagnetized substance



When an external field B_0 is applied, the atomic magnetic moments tends to align with the field, giving the sample a net magnetization.

Some examples of ferromagnetic substance are, iron, cobalt, nickel, gadolinium and dysprosium.

Magnetic Force on a Current – Carrying Conductor

Consider a straight segment of a conducting wire, with length L and cross-sectional area A_i the current is from bottom to top. The wire is in a uniform magnetic field $\vec{B}$, perpendicular to the plane of the diagram and directed into the plane. Let's assume that the moving charges are positive as shown in fig. 1.0 below.

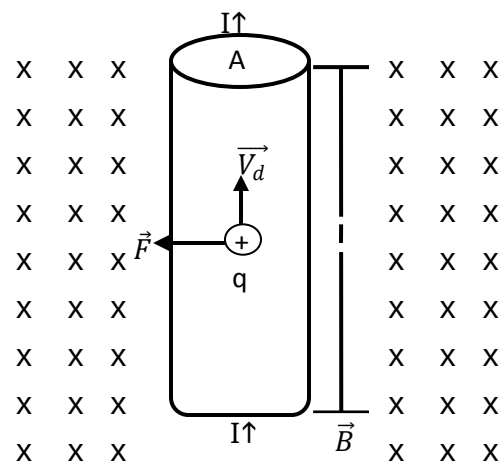


Fig. 1.0: Forces on a moving position charge in a current carrying conductor

The drift velocity $\vec{V}_d$ is upward, perpendicular to $\vec{B}$. The average force on each charge is $\vec{F} = q\vec{V}_d \times \vec{B}$, directed to the left as shown in the figure; since $\vec{V}_d$ and $\vec{B}$ are perpendicular, the magnitude of the force is $F = qV_d B$ (i)

To find the total force on all the moving charges in a length l of conductor with cross-sectional area A , we multiply the force on one charge by the number of charges in the segment. Since the volume of the segment is Al , the number of charges in the segment is nAl , where n is the number of charges per unit volume.

Hence, the total magnetic force on all the moving charges in this segment is

$$F = (qV_d \times B) nAl \quad \text{..... (ii)}$$

But $I = nqV_d A$

Therefore,

$$f = Il \times B \quad \text{..... (iii)}$$

if the $\vec{B}$ field is not perpendicular to the wire but makes an angle θ with it,

$$\Rightarrow f = IlB \sin\theta \quad \text{..... (iv)}$$

If $\theta = 0$ then $f = 0$

If the conductor is not straight, we can divide it into infinite small segments dl . The force df on each segment is

$$df = IdL \times B \quad \dots\dots\dots (v)$$

To get the total force on the wire, we integrated equation (v) over the length of the wire. That is

$$F = I \int_a^b dL \times B \quad \dots\dots\dots (vi)$$

Where a and b represents the end points of the wire

NB Equation (iv) is valid for both positive and negative charges and even when both signs of charge are present at once. This happens in some semiconductor materials and in ionic.

Exercise

Where is the application of the magnetic force on a current-carrying wire found?

- (1) A conductor carrying a current of 150A and 20m in length is placed in a uniform magnetic field of flux density 0.75T. Calculate the force on the conductor if it is placed
 - (i) Perpendicular to the field
 - (ii) at 50° to the field.

Solution

$$(i) \quad F = ILB$$

$$= 150 \times 20 \times 0.75$$

$$= \mathbf{2250N}$$

$$(ii) \quad F = ILB \sin \theta$$

$$= 150 \times 20 \times 0.75 \times \sin 50$$

$$= \mathbf{1723.6N}$$

Exercise

1. A straight horizontal wire X_1 of mass 80g and length 0.6m is placed in a uniform horizontal magnetic field of 0.4T perpendicular to X . Calculate the current in X if the force acting on it just balances its weight ($g = 10\text{N/kg}$)
2. A wire carrying a current of 20A and 4m in length is placed in a field of flux density 0.55T. What is the force on the wire if it is placed:
 - (i) Perpendicular to the field.
 - (ii) Parallel to the field.
 - (iii) At 60° to the field.

- **ELECTROMAGNETIC INDUCTION**
- **D.C. GENERATORS (DYNAMO)**
- **A.C. GENERATORS**

ELECTROMAGNETIC INDUCTION

Classical theory of electromagnetism deals with electric and magnetic fields and interactions caused by distribution of electric charges and currents. It was the work of James Clerk Maxwell which consistently unify the two distinct theories of electricity and magnetism into a single theory called electromagnetism.

Electromagnetic induction is one of the novel discoveries which give birth to classical electrodynamics (electromagnetism). As said earlier, electric field concept in physics i.e. naturally independent.

The theory of electromagnetic induction was first developed by Michael Faraday in 1831. In the 1820's it was known that an electric current produces a magnetic field. Joseph Henry and Michael Faraday both work independently made effort to reverse the process and produce a current with a magnetic field.

The result is elusive because a steady magnetic field will not induce a current to flow in a circuit. It is only a time changing magnetic field that induces a current to flow. Therefore these bodies of experiments was summarized as Faraday's Law of electromagnetic induction.

Electromagnetic Induction is a process where a conductor placed in a changing magnetic field causes the production of E.M.F and electric current.

Thus Faraday's Law of electromagnetic induction state that: whenever there is a change in the magnetic flux linking a circuit on electro motive force (EMF) is induced the strength of which is proportional to the rate of change of the magnetic flux linking the circuit.

$$E = \frac{-d}{dt} \phi_m(t)$$

Where: ϕ_m = magnetic flux

Consider a closed circuit which surface A

$$\phi_m(t) = \int_A \mathbf{x} \cdot \hat{n} \cdot \mathbf{B}_{(t, \mathbf{x})}$$

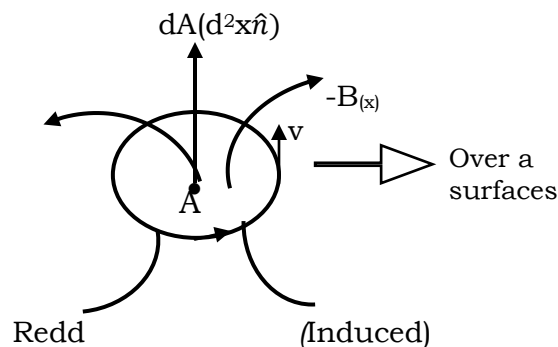


Fig. 1

$B_{(t, \mathbf{x})}$ = magnetic field (induced)

Therefore:

$$E = \frac{-d}{dt} \int_A d^2x \hat{n} \cdot B(t, x)$$

$$= - \int_A d^2x \hat{n} \cdot \frac{2}{4} B(t_1, x)$$

Apply stokes theorem on this integral, transform it to:

$$\nabla \times E_{(t,x)} = \frac{-2}{4} B_{(t,x)} \dots\dots\dots (2)$$

Don't worry much about the equation now:

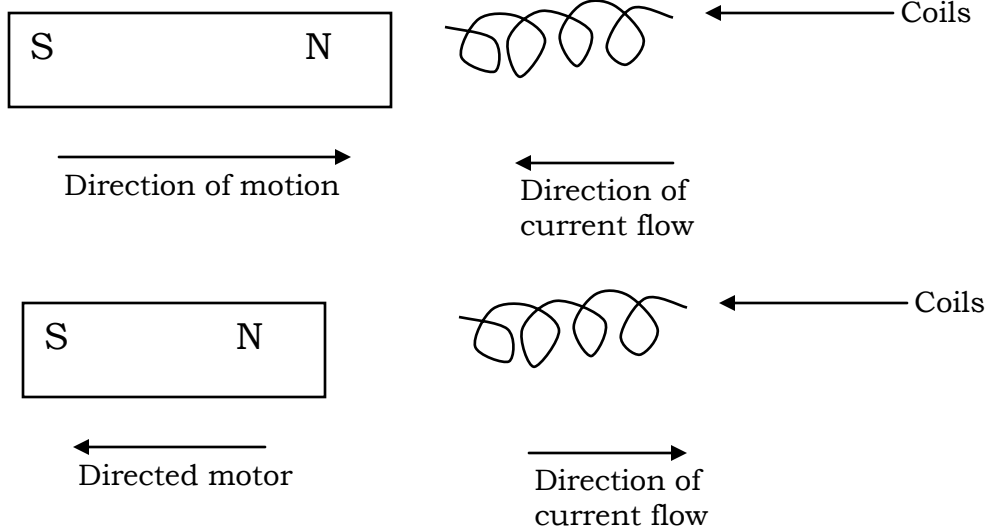
∇ read del, $\times$ is read as cross (note not multiplication symbol) while $\nabla \times E$ is read as curl of electric field.

Note: Equation 2 explicitly connects electricity with magnetism which resulted in technological breakthrough that produces, generators, electric motors, trains, fans, Relays, induction coils and even levitating trains (MAGLEV)

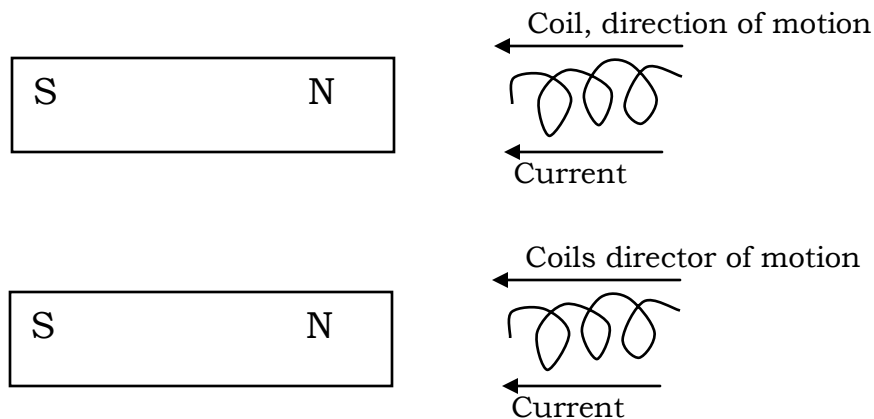
Lenz law

Faraday's law tells us that an EMF is always induced whenever there is a change in magnetic flux linking the circuit or coils either the magnetic field is moving or the coil is moving whichever, but it did not tell us the direction of current flow in the induced E.M.F, this was the success of Lenz called Lenz Law: which states that whenever there is a change in the magnetic flux linking a coil, and induce current is produced which flows in such a way as to oppose the motion producing it (either that of the coil or the magnet). Thus the current in the loop flows in the opposite direction to the motion of either the magnet or the coil.

(a) Moving Magnet



(b) Moving Coil



Lenz Law can be verified in the laboratory using a galvanometer, a bar magnetic and a coil of wire.

DYNAMO (D-C GENERATOR)

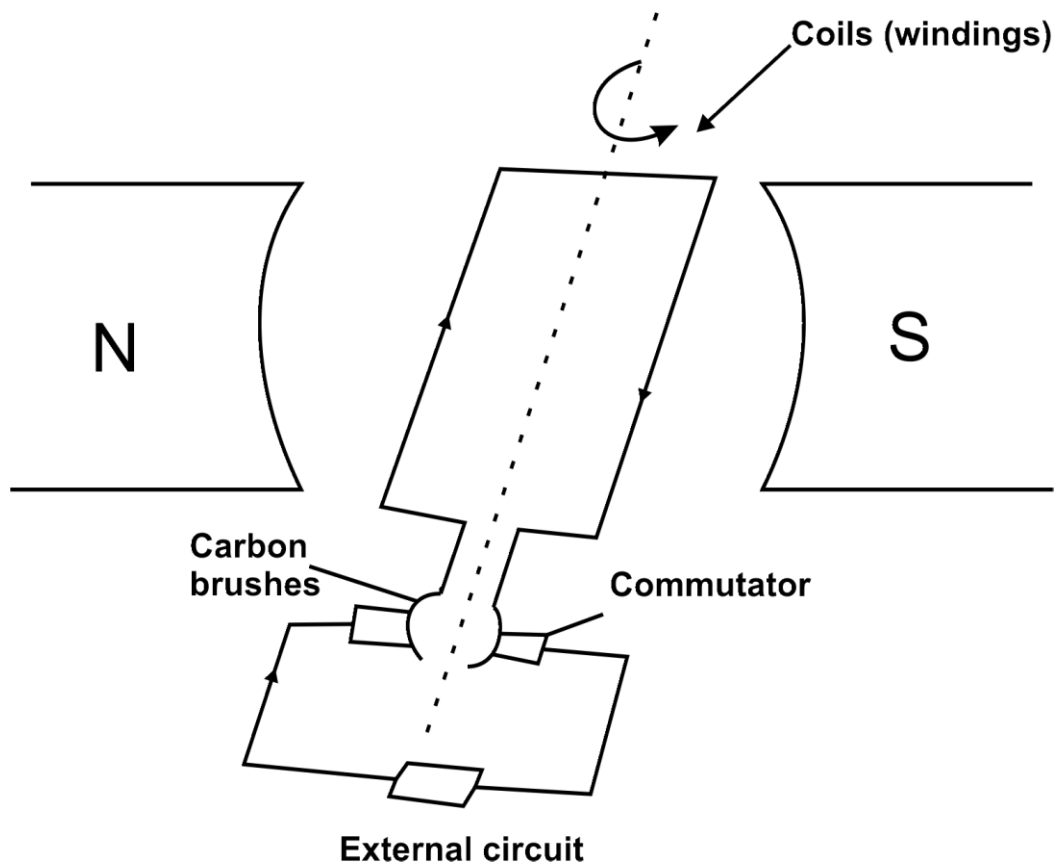
Dynamo is an electrical generator that produces direct current using a commutator. Dynamo is most often called D.C. generator. Dynamos were the first electrical generators capable of delivering

power for industrial usage. Though A.C. generators have a better advantage over D.C. generator, no doubt Dynamo is a foundation upon which other electric-power conversion machine were based. Dynamos were invented as a viable replacement for batteries. The use of A.C. current is not known then.

Principles

A dynamic machine consists of a stationary structure called stator, which provides a constant magnetic held and a set of rotating windings called armature which turn within the constant magnetic field; cutting the magnetic flux thereby creating an electric current in the wire. A commutator was needed to produce direct current because when a loop of wire rotate in a magnetic field the current produce reverses with each half turn which produces alternating current, the commutator serve as a rotary switch, it consist of a set of contacts mounted on the machines shaft combined with graphite-block station by contacts called “brushes”. The commutator reverses the connection of the windings to the external circuit when the potential reverses, therefore instead of alternating current, a direct current is produced.

Note: a D.C. generator in reverse order could work as a D.C. motor.



A.C. GENERATOR

A.C. generators or alternators as they are usually called. A.C. generators produce alternating current with frequency base on the rotational speed of the rotor and the number of magnetic poles.

Principles

An A.C. generator is the same with D.C generator except A.C generator does not have a commentator. A typical alternators use a rotating field (electro magnetic) winding excited with a direct current and a stationary winding (stator) to produce alternating current.

Generators consist mainly of two parts with subdivision namely the mechanical parts, and the electrical parts.

Mechanical parts

- (1) **Rotor:** Wheel is the rotating part of an electrical machine.
- (2) **Stator:** Which is the stationary part of an electrical machine.

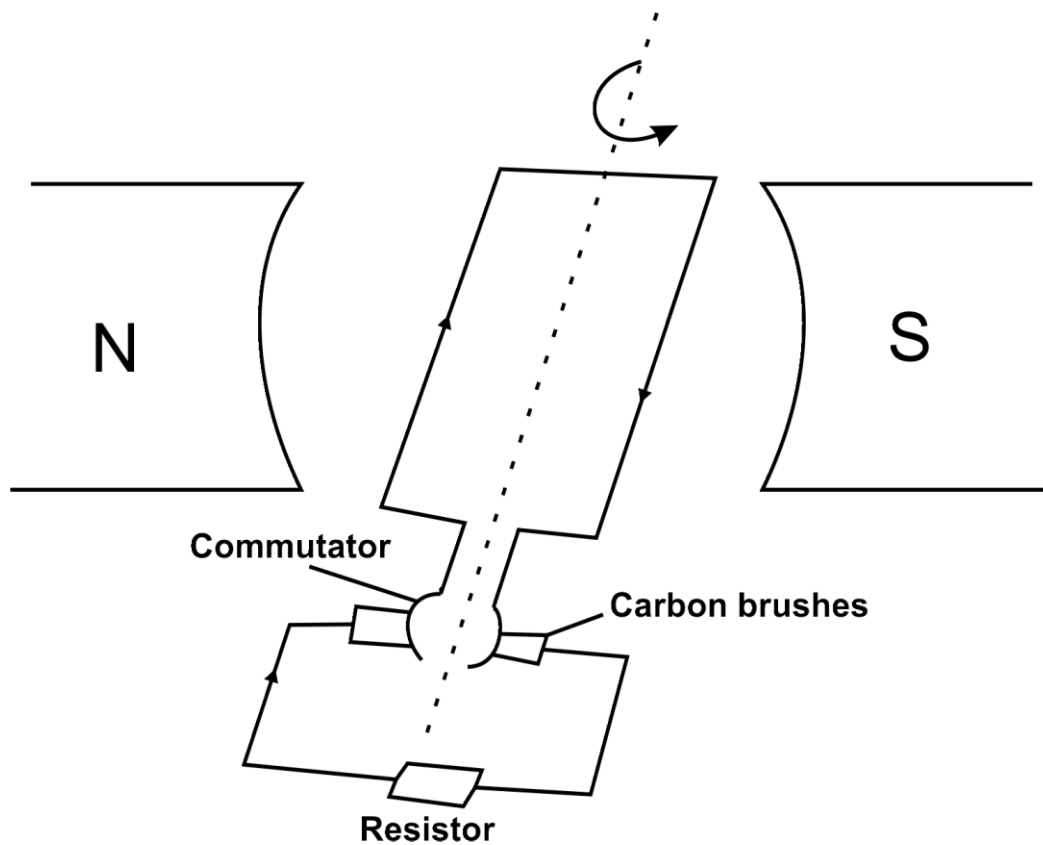
Armature: Which is the windings that generate the electric Current.

Electrical Parts

- 1. Field

Magnetic Excitation of Electric Generators

Generators that use field coils, rather than permanent magnet requires a current to be present in the field coils for the device to work; this is known as excitation. A small generator is self-excited that is when the generator first start to turn, the small amount of magnetism present in the iron core provides a magnetic field which gets the generator started, generating a current also in the armature. Large generators like the one use in power stations require a separate smaller generator to excite the field coils to input large armature current to start it.



Note: A.C. generators also function as A.C. motors in reverse order.